

CAPESTONE PROJECT:

Scenario1: Automate Orange HRM login and logout with 5 different data set including valid and invalid data set.

- 1. Save data set in Excel file**
- 2. Write a script to read data from excel**
- 3. Prepare script for Login and logout**
- 4. Perform assertion for valid data set (use Username: Admin and Password: admin123) test case should be pass and invalid data set (other than given data) test case should be fail.**
- 5. Capture Screenshot for every login functionality**
- 6. Generate Extent Report for the same.**

CODE:

`ExcelReader.java`

```
package scenario1;
```

```
import java.io.File;
```

```
import java.io.FileInputStream;
```

```
import java.io.FileNotFoundException;
```

```
import java.io.IOException;
```

```
import org.apache.poi.xssf.usermodel.*;
```

```
import org.testng.annotations.Test;
```

```
public class ExcelReader
```

```
{
```

```
    @Test
```

```
    public void dataRead() throws IOException
```

```
{
```

```
    //File
```

```
    File f1=new File(System.getProperty("user.dir")+"//TestData//Data.xlsx");
```

```
    FileInputStream fs=new FileInputStream(f1);
```

```
    //Workbook
```

```
    XSSFWorkbook wb=new XSSFWorkbook(fs);
```

```

//Number of rows
int rows=wb.getSheet("userdata").getPhysicalNumberOfRows();//5

//Number of Cells
int cells=wb.getSheet("userdata").getRow(0).getPhysicalNumberOfCells();//2

//Array of same size like file
Object data[][]=new Object[rows-1][cells];

for(int i=1;i<rows;i++)//rows
{
    for(int j=0;j<cells;j++)
    {
        data[i-
1][j]=wb.getSheet("userdata").getRow(i).getCell(j).getStringCellValue();
        System.out.print(data[i-1][j]+" ");
    }

    System.out.println();
}
}
}

```

DataDrivenTest.java

```

package scenario1;

import java.io.File;
import java.io.IOException;
import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.OutputType;
import org.openqa.selenium.TakesScreenshot;
import org.openqa.selenium.WebDriver;

```

```

import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.io.FileHandler;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.Test;

public class DataDrivenTest
{

    public static WebDriver driver;
    @Test(dataProvider = "Excel",dataProviderClass = CustomData.class)
    public void testdatadriven(String username,String password) throws
InterruptedException {
        driver = new ChromeDriver();
        //driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
        driver.get("https://opensource-
demo.orangehrmlive.com/web/index.php/auth/login");

        driver.findElement(By.name("username")).sendKeys(username);
        driver.findElement(By.name("password")).sendKeys(password);
        driver.findElement(By.xpath("//button[@type='submit']")).click();

        Thread.sleep(2000);

        Assert.assertTrue(driver.getCurrentUrl().contains("dashboard"),"Login Fail");
        System.out.println("Login Successfull");

        // driver.quit();

    }

    public void captureScreenshot(Object fileName)
    {
        TakesScreenshot ts=(TakesScreenshot)driver;
        File src=ts.getScreenshotAs(OutputType.FILE);
    }
}

```

```

        File dest=new
File(System.getProperty("user.dir")+ "//Screenshots//"+fileName);
        try {
            FileHandler.copy(src, dest);
        } catch (IOException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        }
        System.out.println("ScreenShot captured!");
    }
}

```

CustomData.java

```
package scenario1;
```

```

import java.io.File;
import java.io.FileInputStream;
import java.io.FileNotFoundException;
import java.io.IOException;

```

```

import org.apache.poi.xssf.usermodel.XSSFSheet;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;
import org.testng.annotations.DataProvider;

```

```

public class CustomData
{

```

```
    public XSSFWorkbook wb;
```

```
    @DataProvider(name = "Excel")
```

```
    public Object[][] excelData() {
```

```

        File f1 = new File(System.getProperty("user.dir") +
"//TestData//Data.xlsx");

```

```
        FileInputStream fis;
```

```
        Object arr[][] = null;
```

```
        try {
```

```

        fis = new FileInputStream(f1);
        wb = new XSSFWorkbook(fis);
        XSSFSheet sheet = wb.getSheet("userdata");
        // Number of rows
        int row = sheet.getPhysicalNumberOfRows();
        int cell = sheet.getRow(0).getPhysicalNumberOfCells();

        arr = new Object[row][cell];

        for (int i = 0; i < row; i++) {
            for (int j = 0; j < cell; j++) {
                arr[i ][j] =
sheet.getRow(i).getCell(j).getStringCellValue();

            }

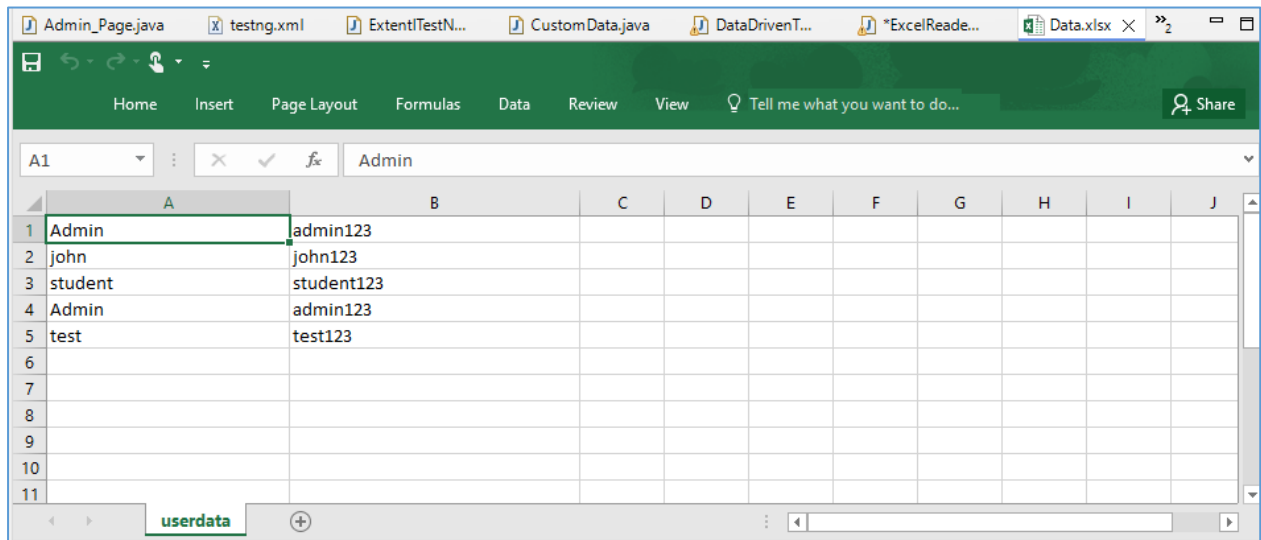
        }
    } catch (FileNotFoundException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    } catch (IOException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    }

    return arr;

}
}

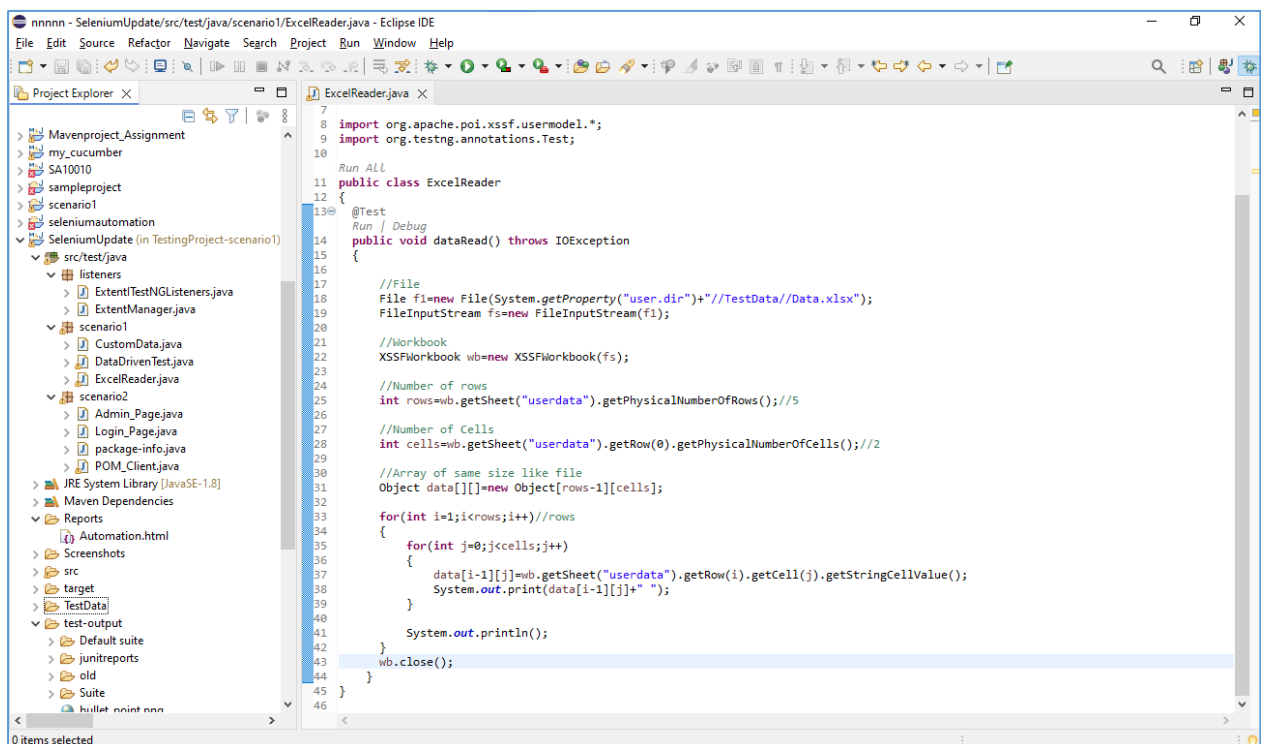
```

1. Save data set in Excel file



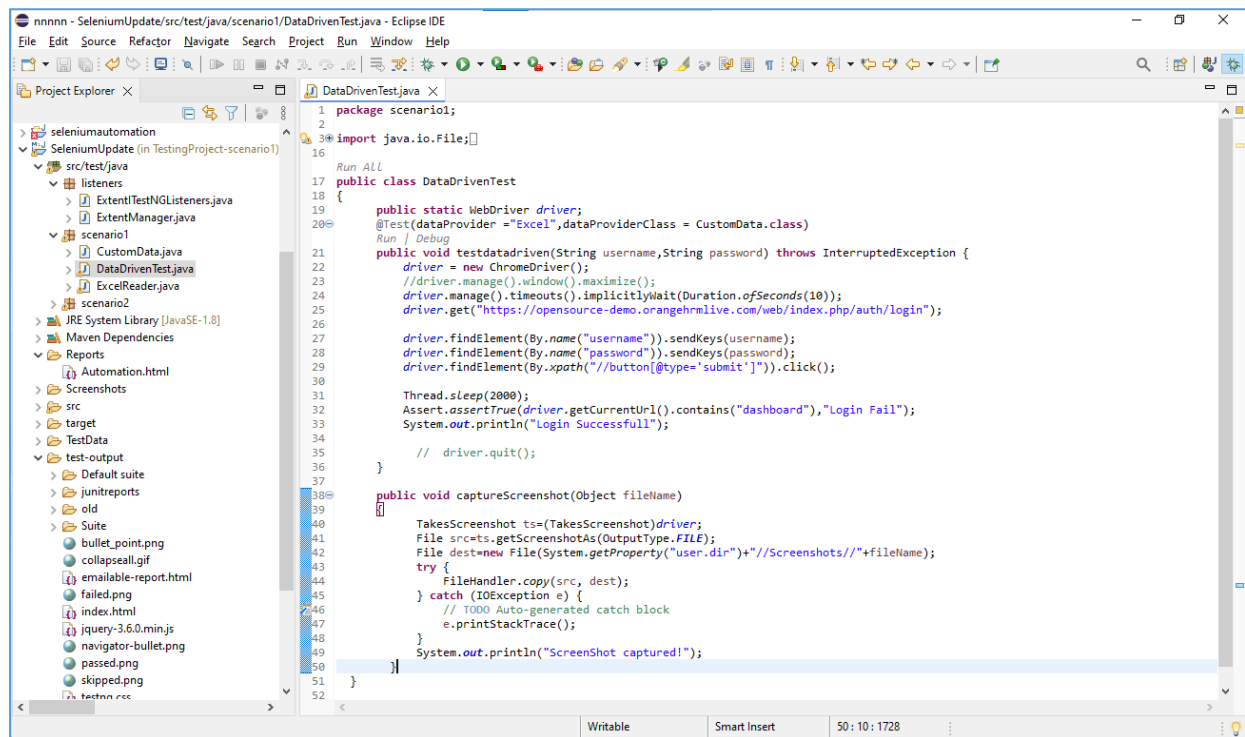
	A	B	C	D	E	F	G	H	I	J
1	Admin	admin123								
2	john	john123								
3	student	student123								
4	Admin	admin123								
5	test	test123								
6										
7										
8										
9										
10										
11										

2. Write a script to read data from excel

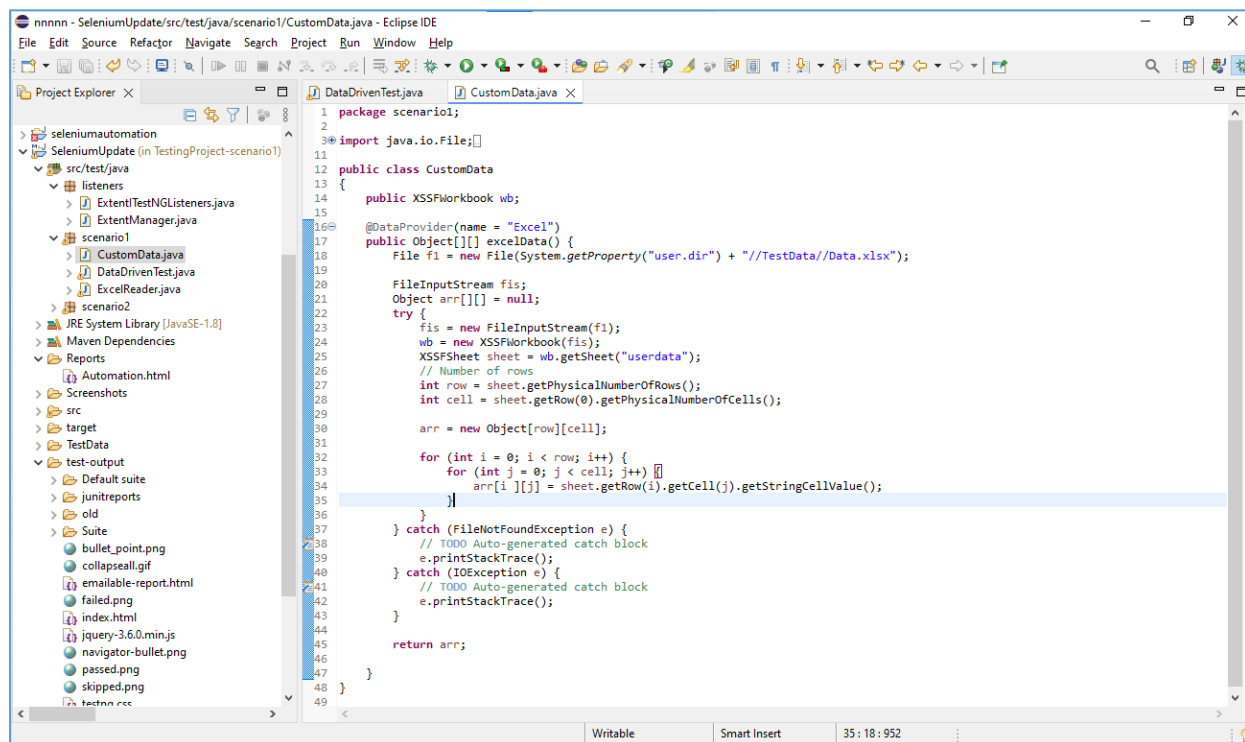


```
7 import org.apache.poi.xssf.usermodel.*;
8 import org.testng.annotations.Test;
9
10
11 Run ALL
12 public class ExcelReader
13 {
14     @Test
15     Run / Debug
16     public void dataRead() throws IOException
17     {
18         //File
19         File f1=new File(System.getProperty("user.dir")+"//TestData//Data.xlsx");
20         FileInputStream fs=new FileInputStream(f1);
21
22         //Workbook
23         XSSFWorkbook wb=new XSSFWorkbook(fs);
24
25         //Number of rows
26         int rows=wb.getSheet("userdata").getPhysicalNumberOfRows();//5
27
28         //Number of Cells
29         int cells=wb.getSheet("userdata").getRow(0).getPhysicalNumberOfCells();//2
30
31         //Array of same size like file
32         Object data[][]=new Object[rows-1][cells];
33
34         for(int i=1;i<rows;i++)//rows
35         {
36             for(int j=0;j<cells;j++)
37             {
38                 data[i-1][j]=wb.getSheet("userdata").getRow(i).getCell(j).getStringCellValue();
39                 System.out.print(data[i-1][j]+" ");
40             }
41             System.out.println();
42         }
43         wb.close();
44     }
45 }
46
```

3. Prepare script for Login and logout

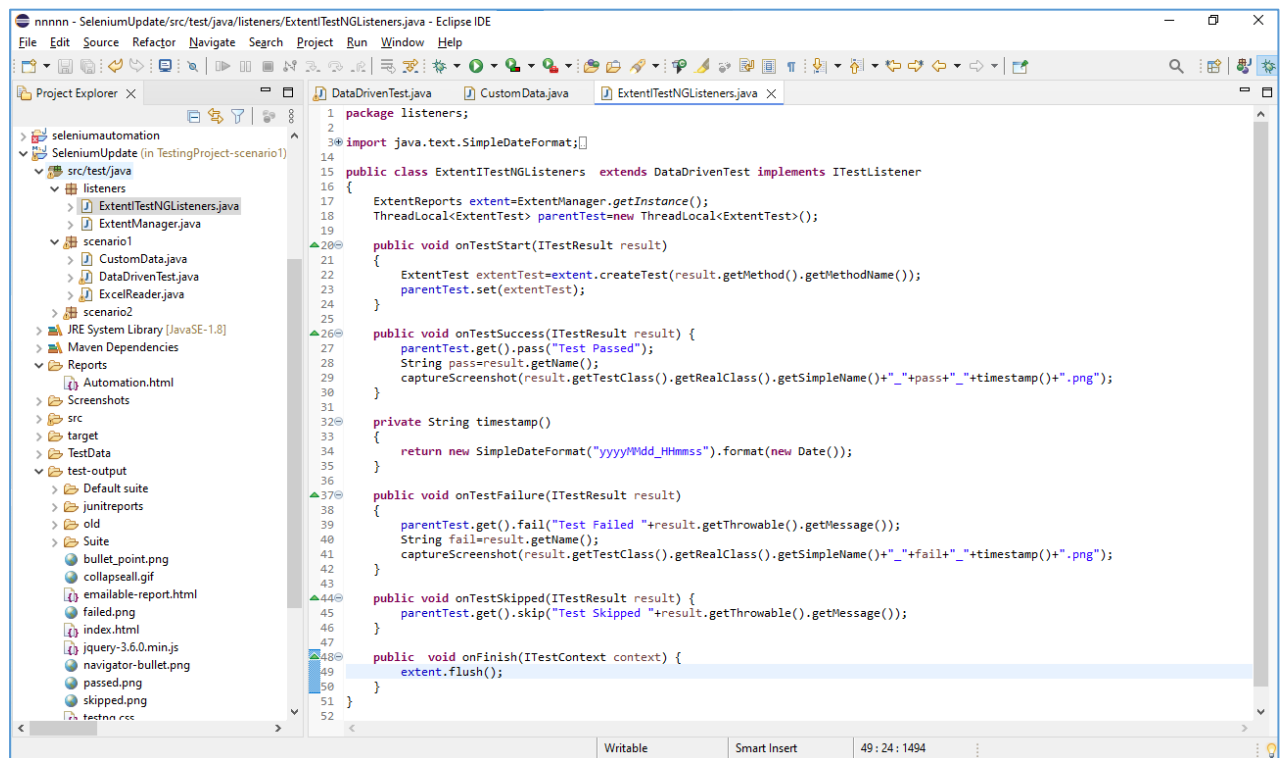


```
1 package scenariol;
2
3 import java.io.File;
4
5 Run ALL
6
7 public class DataDrivenTest
8 {
9     public static WebDriver driver;
10    @Test(dataProvider = "Excel",dataProviderClass = CustomData.class)
11    Run | Debug
12    public void testdatadriven(String username,String password) throws InterruptedException {
13        driver = new ChromeDriver();
14        //driver.manage().window().maximize();
15        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
16        driver.get("https://opensource-demo.orangehrmlive.com/web/index.php/auth/login");
17
18        driver.findElement(By.name("username")).sendKeys(username);
19        driver.findElement(By.name("password")).sendKeys(password);
20        driver.findElement(By.xpath("//button[@type='submit']")).click();
21
22        Thread.sleep(2000);
23        Assert.assertTrue(driver.getCurrentUrl().contains("dashboard"),"Login Fail");
24        System.out.println("Login Successful");
25
26        // driver.quit();
27    }
28
29    public void captureScreenshot(Object fileName)
30    {
31        TakesScreenshot ts=(TakesScreenshot)driver;
32        File src=ts.getScreenshotAs(OutputType.FILE);
33        File dest=new File(System.getProperty("user.dir")+"//Screenshots/"+fileName);
34        try {
35            FileHandler.copy(src, dest);
36        } catch (IOException e) {
37            // TODO Auto-generated catch block
38            e.printStackTrace();
39        }
40        System.out.println("ScreenShot captured!");
41    }
42 }
```



```
1 package scenariol;
2
3 import java.io.File;
4
5 public class CustomData
6 {
7     public XSSFWorkbook wb;
8
9     @DataProvider(name = "Excel")
10    public Object[][] excelData() {
11        File f1 = new File(System.getProperty("user.dir") + "//TestData/Data.xlsx");
12
13        FileInputStream fis;
14        Object arr[][] = null;
15        try {
16            fis = new FileInputStream(f1);
17            wb = new XSSFWorkbook(fis);
18            XSSFSheet sheet = wb.getSheet("userdata");
19            // Number of rows
20            int row = sheet.getPhysicalNumberOfRows();
21            int cell = sheet.getRow(0).getPhysicalNumberOfCells();
22
23            arr = new Object[row][cell];
24
25            for (int i = 0; i < row; i++) {
26                for (int j = 0; j < cell; j++) {
27                    arr[i][j] = sheet.getRow(i).getCell(j).getStringCellValue();
28                }
29            }
30        } catch (FileNotFoundException e) {
31            // TODO Auto-generated catch block
32            e.printStackTrace();
33        } catch (IOException e) {
34            // TODO Auto-generated catch block
35            e.printStackTrace();
36        }
37
38        return arr;
39    }
40 }
```

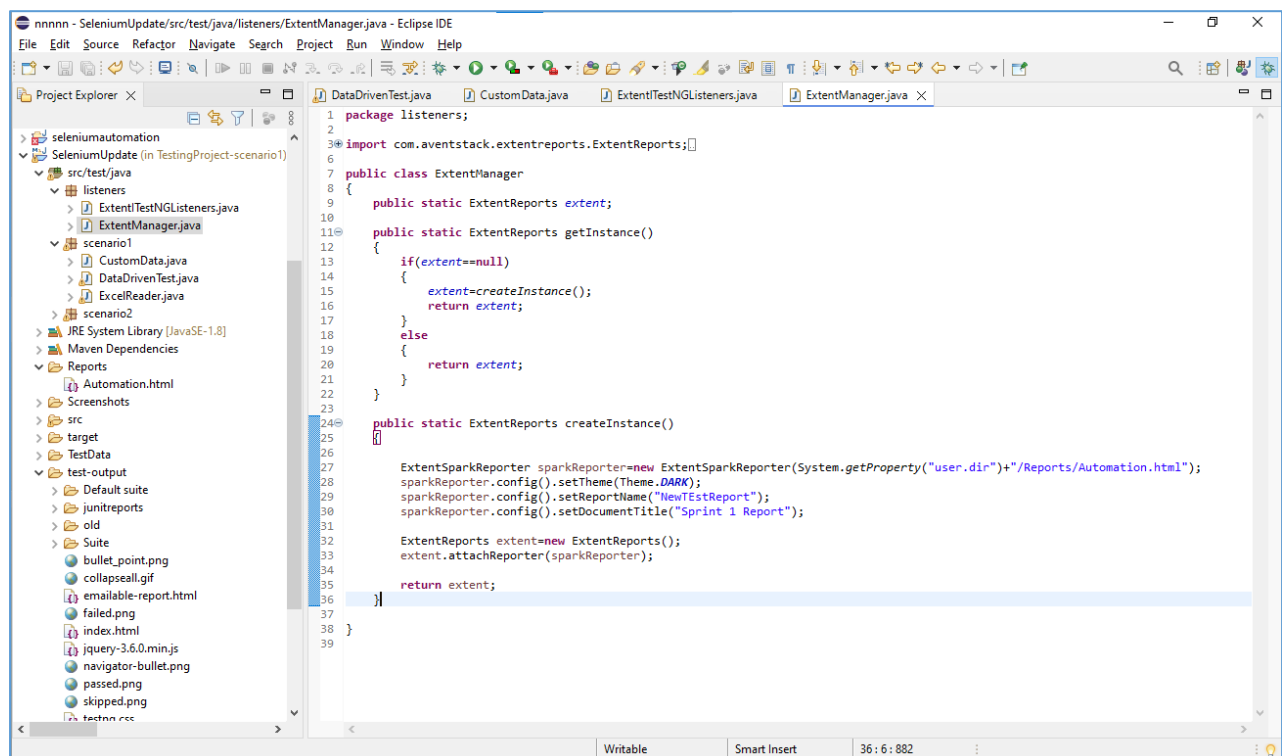
4. Perform assertion for valid data set (use Username: Admin and Password: admin123) test case should be pass and invalid data set (other than given data) test case should be fail.



The screenshot shows the Eclipse IDE with the file `ExtentTestNGListeners.java` open. The code implements the `IListener` interface from TestNG. It includes methods for `onTestStart`, `onTestSuccess`, `onTestFailure`, `onTestSkipped`, and `onFinish`. The `onTestSuccess` method captures a screenshot and logs the test name, class name, and timestamp. The `onTestFailure` method also captures a screenshot and logs the failure message. The `onFinish` method flushes the ExtentReports object.

```
1 package listeners;
2
3 import java.text.SimpleDateFormat;
4
5 public class ExtentTestNGListeners extends DataDrivenTest implements IListener
6 {
7     ExtentReports extent=ExtentManager.getInstance();
8     ThreadLocal<ExtentTest> parentTest=new ThreadLocal<ExtentTest>();
9
10    public void onTestStart(ITestResult result)
11    {
12        ExtentTest extentTest=extent.createTest(result.getMethod().getMethodName());
13        parentTest.set(extentTest);
14    }
15
16    public void onTestSuccess(ITestResult result) {
17        parentTest.get().pass("Test Passed");
18        String pass=result.getName();
19        captureScreenshot(result.getTestClass().getRealClass().getSimpleName()+"_"+pass+"_"+timestamp()+".png");
20    }
21
22    private String timestamp()
23    {
24        return new SimpleDateFormat("yyyyMMdd_HHmss").format(new Date());
25    }
26
27    public void onTestFailure(ITestResult result)
28    {
29        parentTest.get().fail("Test Failed "+result.getThrowable().getMessage());
30        String fail=result.getName();
31        captureScreenshot(result.getTestClass().getRealClass().getSimpleName()+"_"+fail+"_"+timestamp()+".png");
32    }
33
34    public void onTestSkipped(ITestResult result) {
35        parentTest.get().skip("Test Skipped "+result.getThrowable().getMessage());
36    }
37
38    public void onFinish(ITestContext context) {
39        extent.flush();
40    }
41 }
```

5. Capture Screenshot/Extent report for every login functionality



The screenshot shows the Eclipse IDE with the file `ExtentManager.java` open. The code defines the `ExtentManager` class, which is responsible for creating and managing the `ExtentReports` object. It includes a `getInstance` method that returns a singleton instance of `ExtentReports` and a `createInstance` method that creates a new instance of `ExtentReports` and attaches a `SparkReporter` to it.

```
1 package listeners;
2
3 import com.aventstack.extentreports.ExtentReports;
4
5 public class ExtentManager
6 {
7     public static ExtentReports extent;
8
9     public static ExtentReports getInstance()
10    {
11        if(extent==null)
12        {
13            extent=createInstance();
14            return extent;
15        }
16        else
17        {
18            return extent;
19        }
20    }
21
22    public static ExtentReports createInstance()
23    {
24        ExtentSparkReporter sparkReporter=new ExtentSparkReporter(System.getProperty("user.dir")+"/Reports/Automation.html");
25        sparkReporter.config().setTheme(Theme.DARK);
26        sparkReporter.config().setReportName("NewTestReport");
27        sparkReporter.config().setDocumentTitle("Sprint 1 Report");
28
29        ExtentReports extent=new ExtentReports();
30        extent.attachReporter(sparkReporter);
31
32        return extent;
33    }
34 }
```

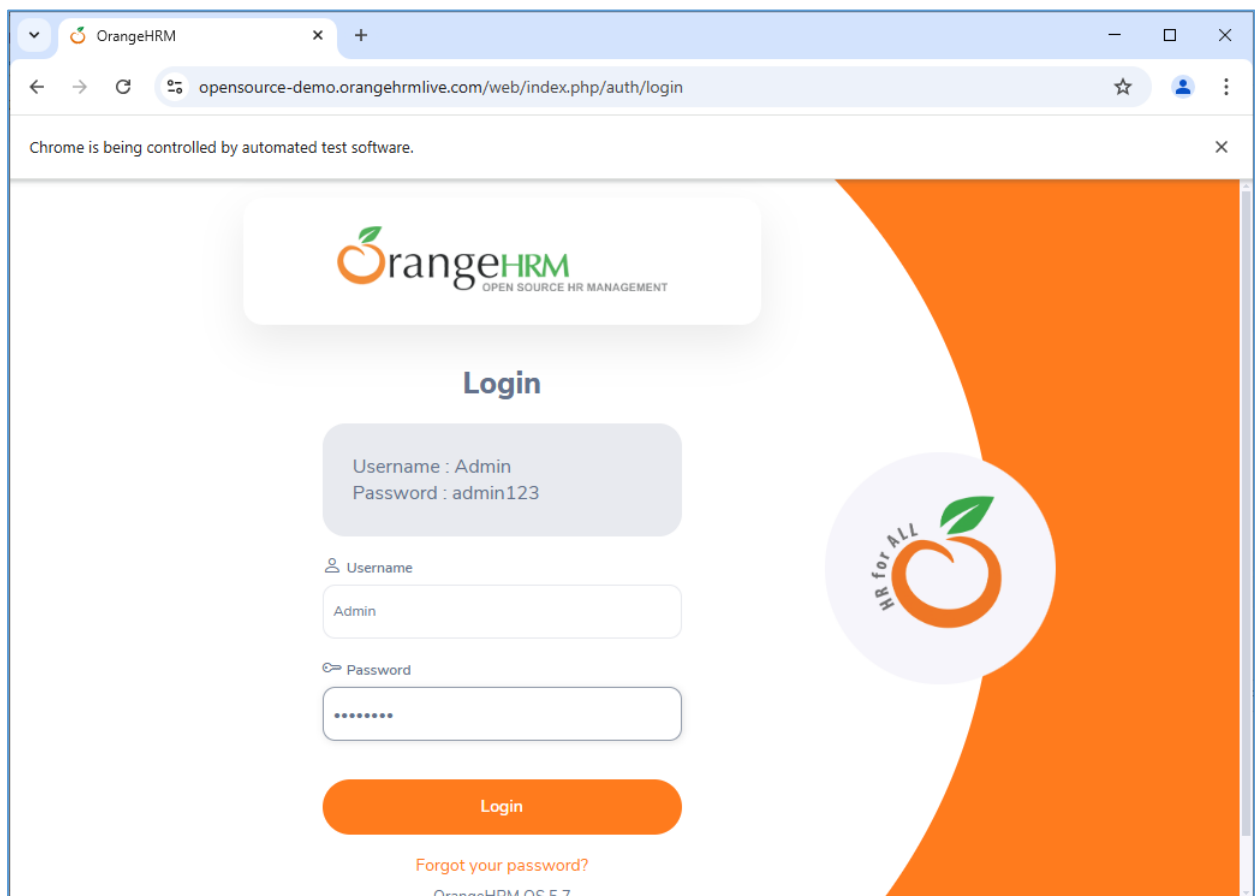

CONSOLE OUTPUT:

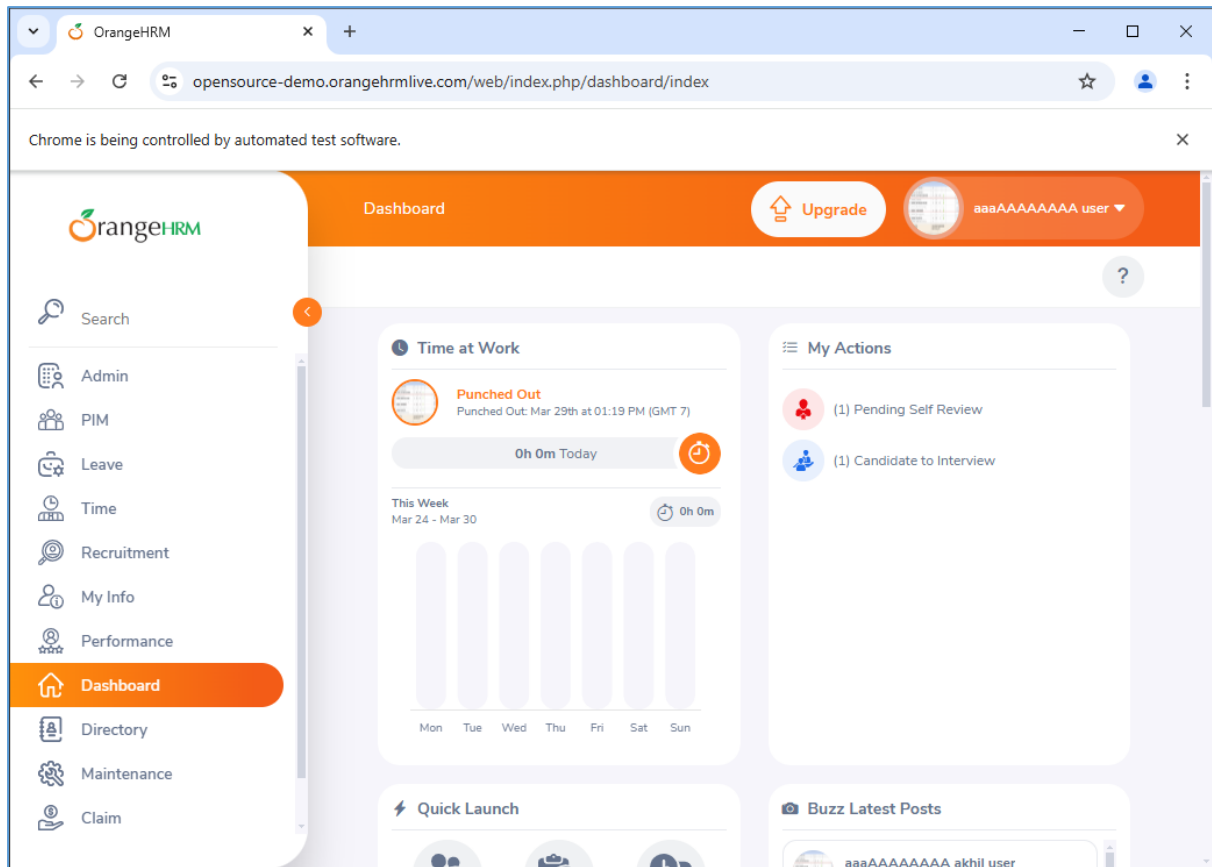
```
nnnnn - SeleniumUpdate/pom.xml - Eclipse IDE
File Edit Navigate Search Project Run Window Help

[RemoteTesting] detected TestNG version 7.7.1
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
ERROR StatusLogger Log4j2 could not find a logging implementation. Please add log4j-core to the classpath. Using SimpleLogger to log to the console...
Mar 30, 2025 4:28:40 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
Mar 30, 2025 4:28:40 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$5
Login Successful
Screenshot captured!
Mar 30, 2025 4:28:59 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 134
Mar 30, 2025 4:28:59 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$5
WARNING: Unable to find version of CDP to use for 134.0.6998.178. You may need to include a dependency on a specific version of the CDP using something similar to `org.selenium
Login Fail
Screenshot captured!
Mar 30, 2025 4:29:12 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 134
Mar 30, 2025 4:29:12 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$5
WARNING: Unable to find version of CDP to use for 134.0.6998.178. You may need to include a dependency on a specific version of the CDP using something similar to `org.selenium
Login Fail
Screenshot captured!
Mar 30, 2025 4:29:29 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 134
Mar 30, 2025 4:29:29 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$5
WARNING: Unable to find version of CDP to use for 134.0.6998.178. You may need to include a dependency on a specific version of the CDP using something similar to `org.selenium
Login Successful
Screenshot captured!
Mar 30, 2025 4:30:05 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 134
Mar 30, 2025 4:30:05 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$5
WARNING: Unable to find version of CDP to use for 134.0.6998.178. You may need to include a dependency on a specific version of the CDP using something similar to `org.selenium
Login Fail
Screenshot captured!

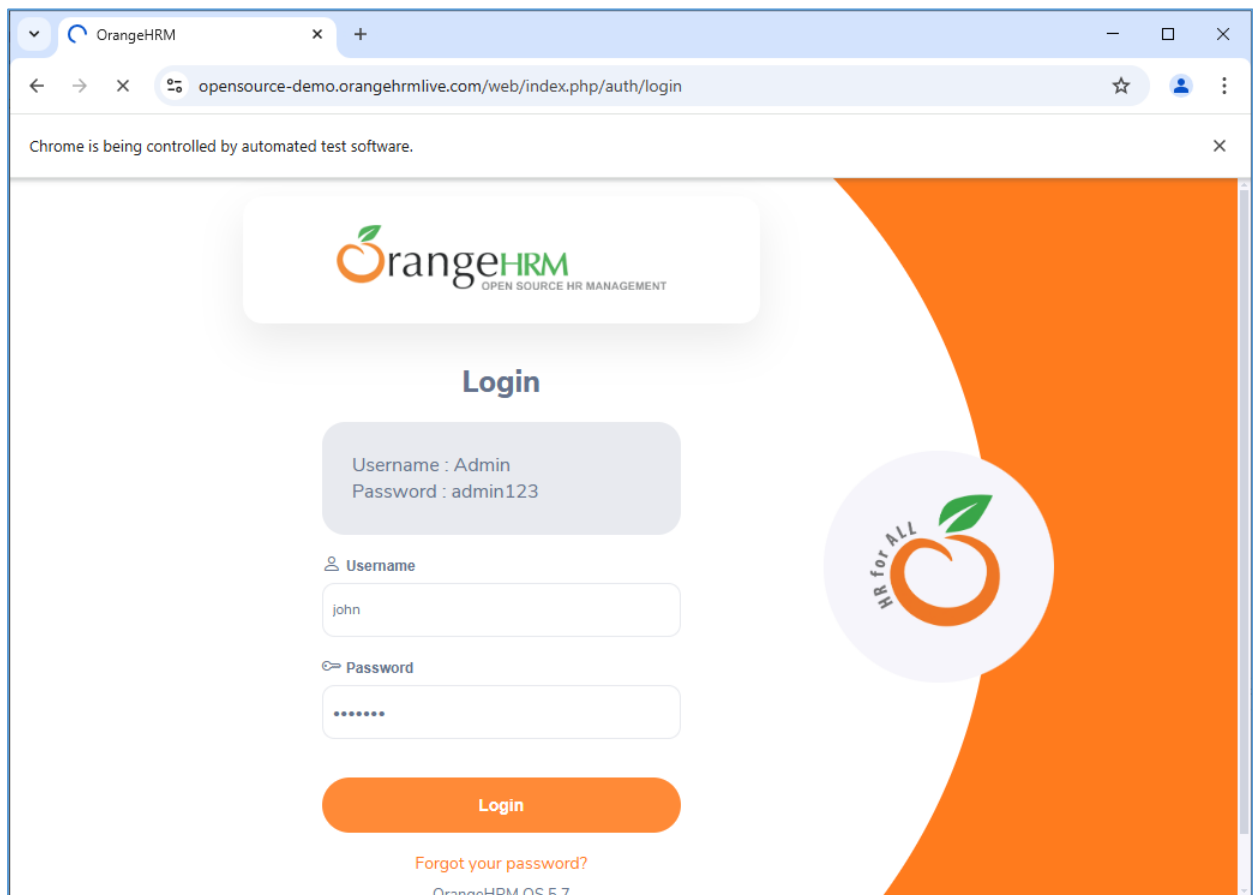
=====
Suite
Total tests run: 5, Passes: 5, Failures: 0, Skips: 0
=====
```

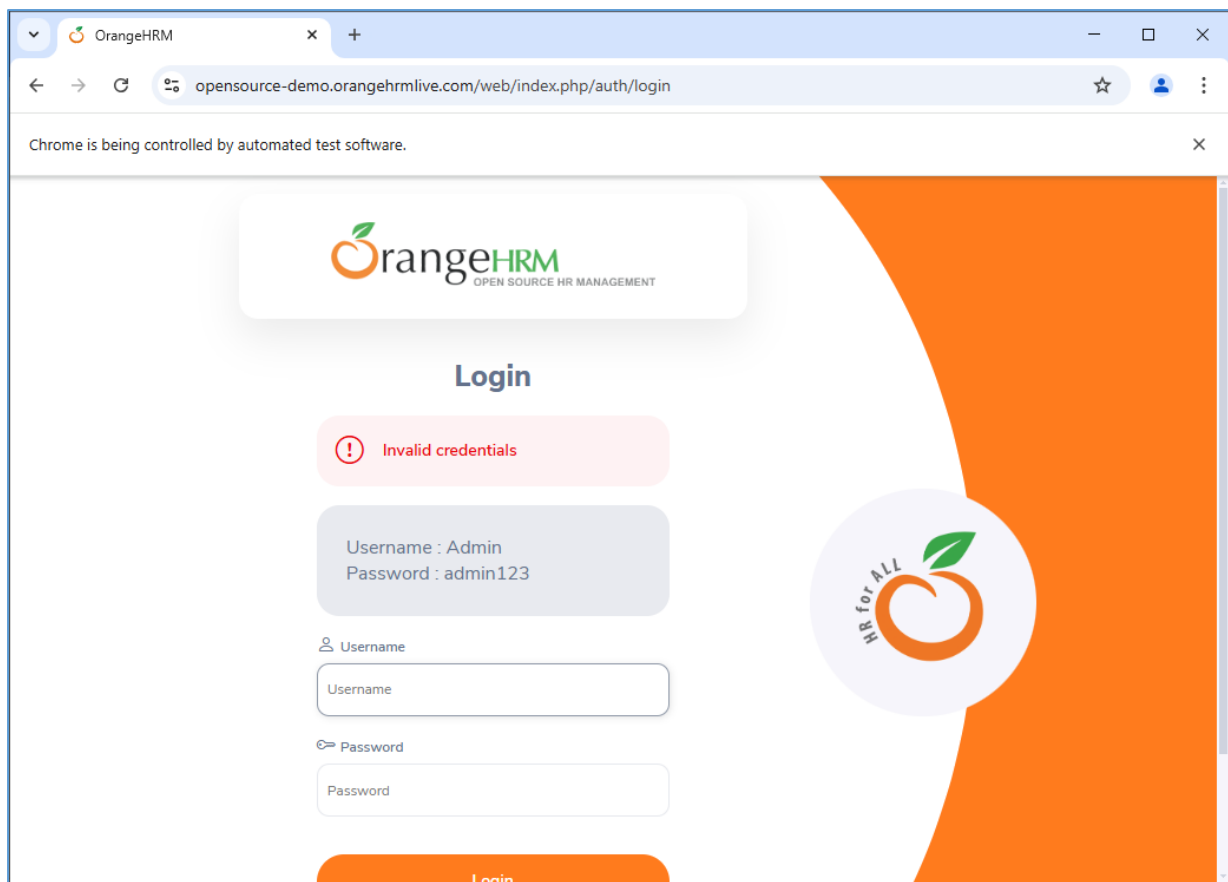
LOGIN 1:



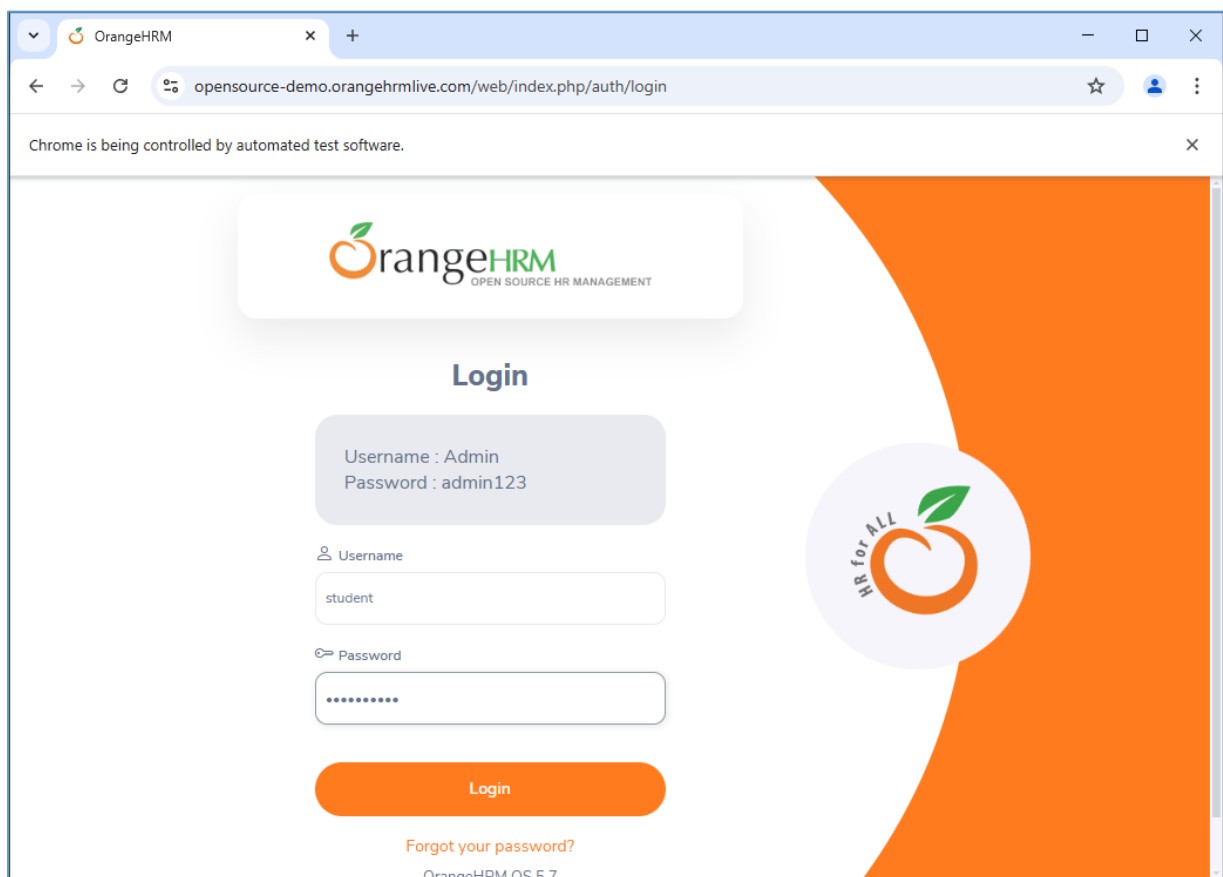


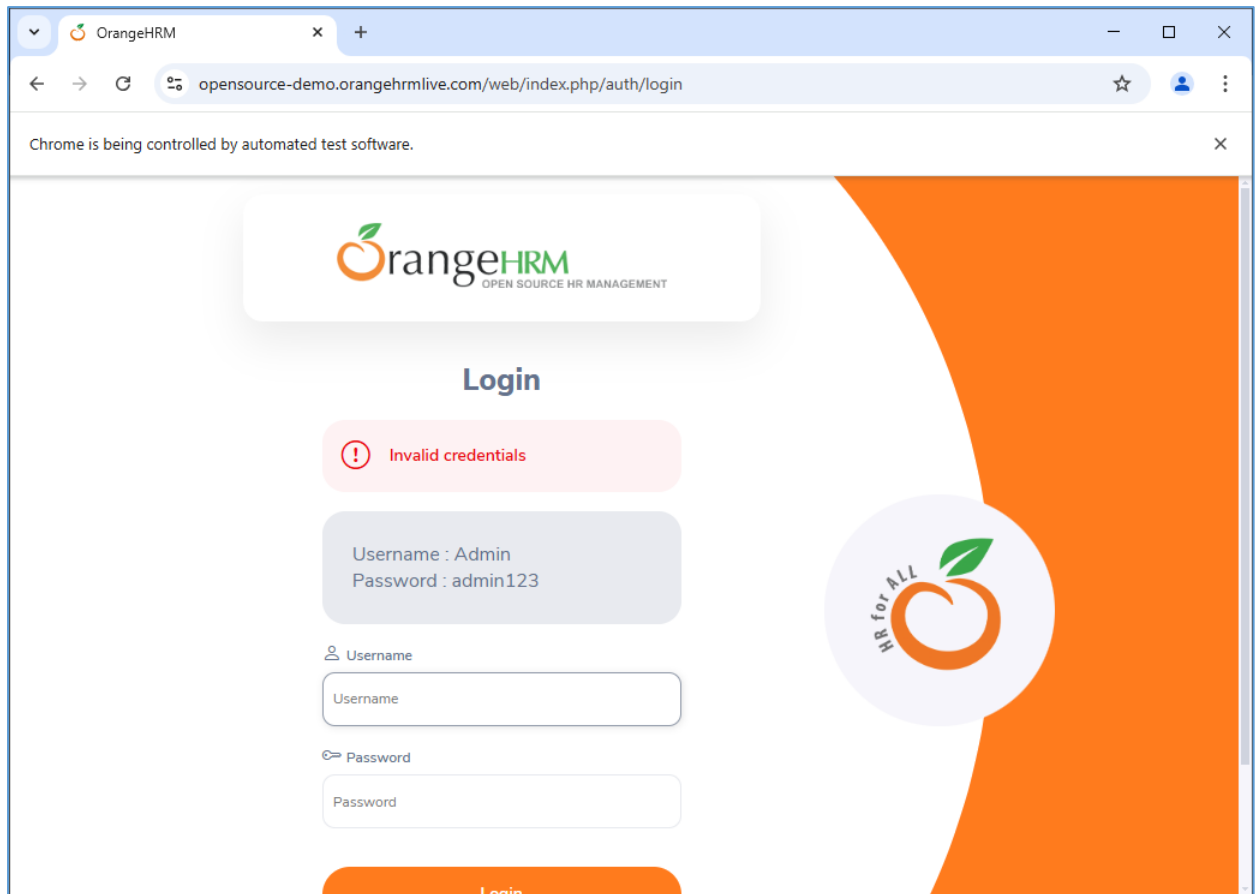
LOGIN2:



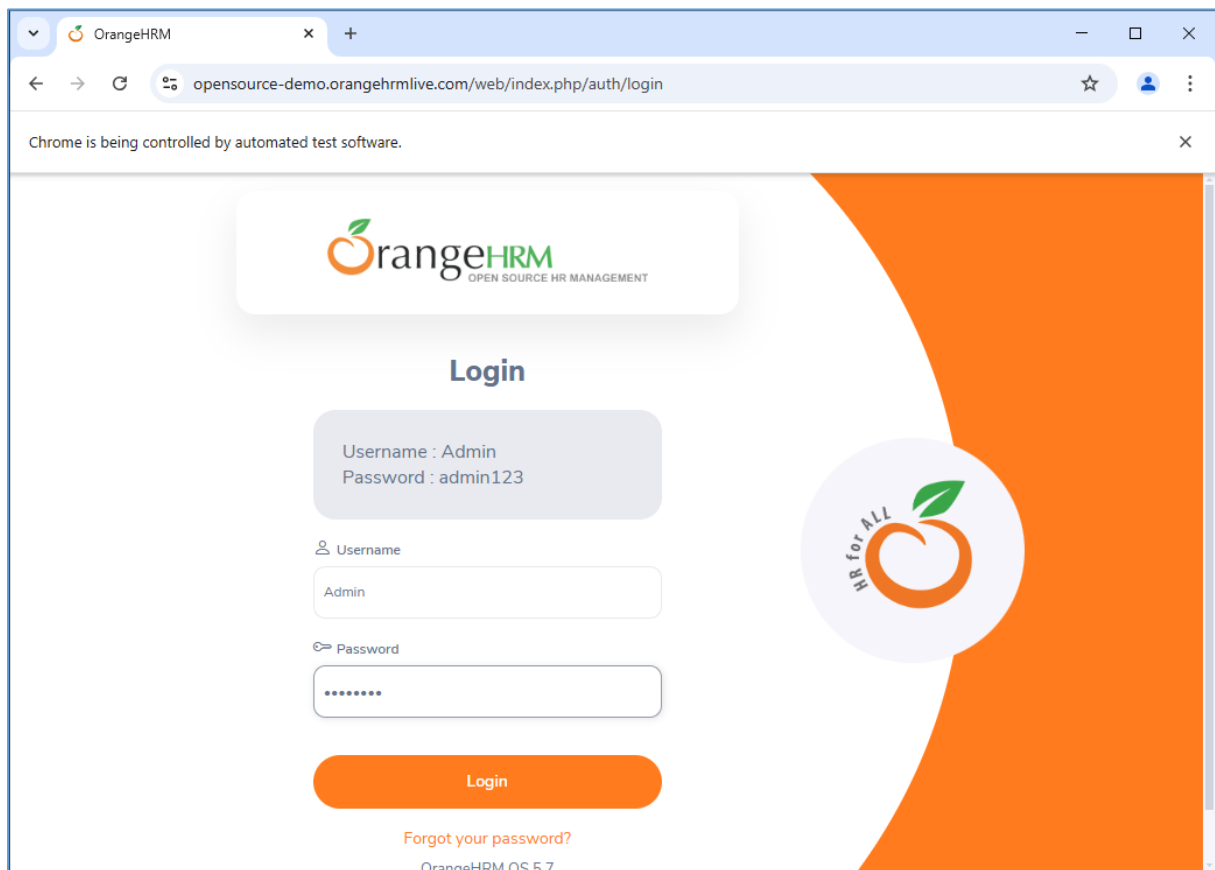


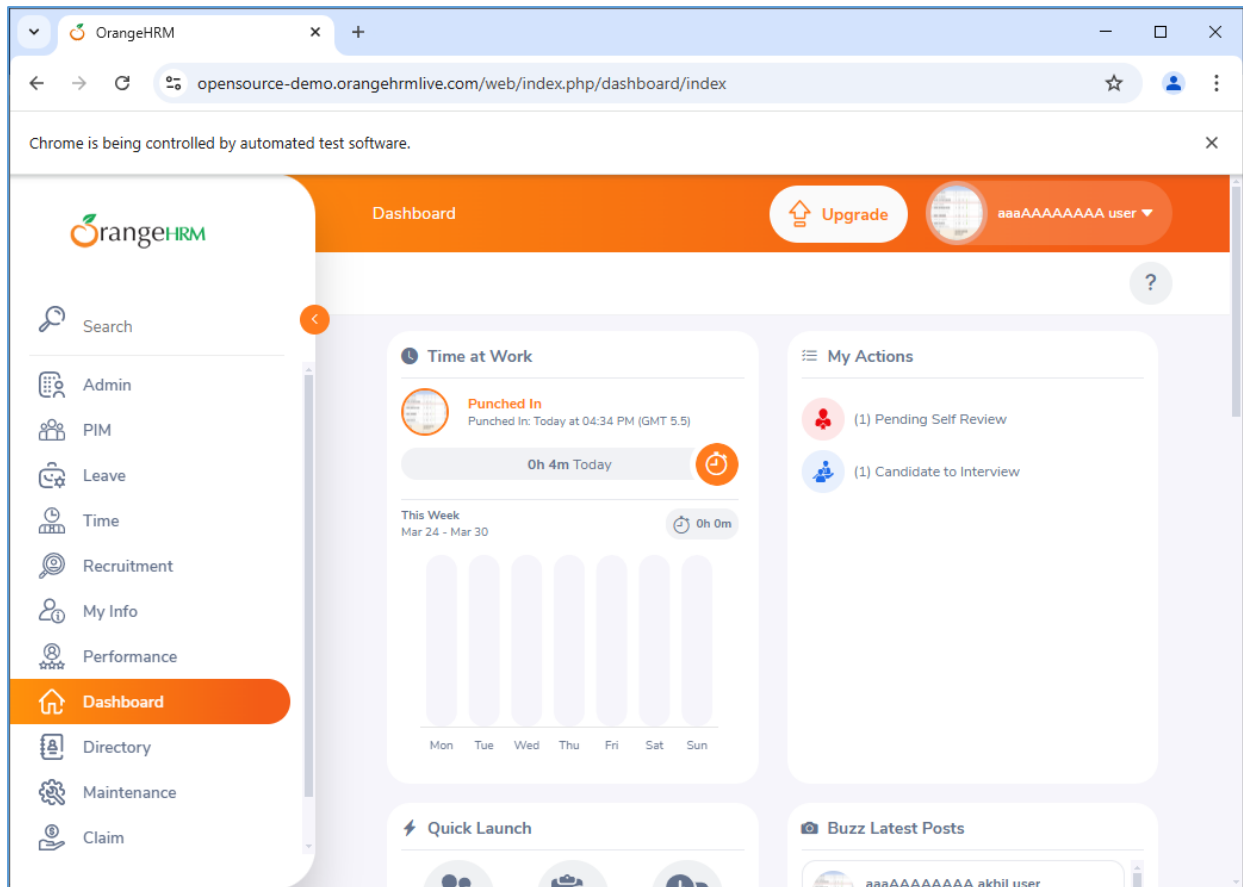
LOGIN3:



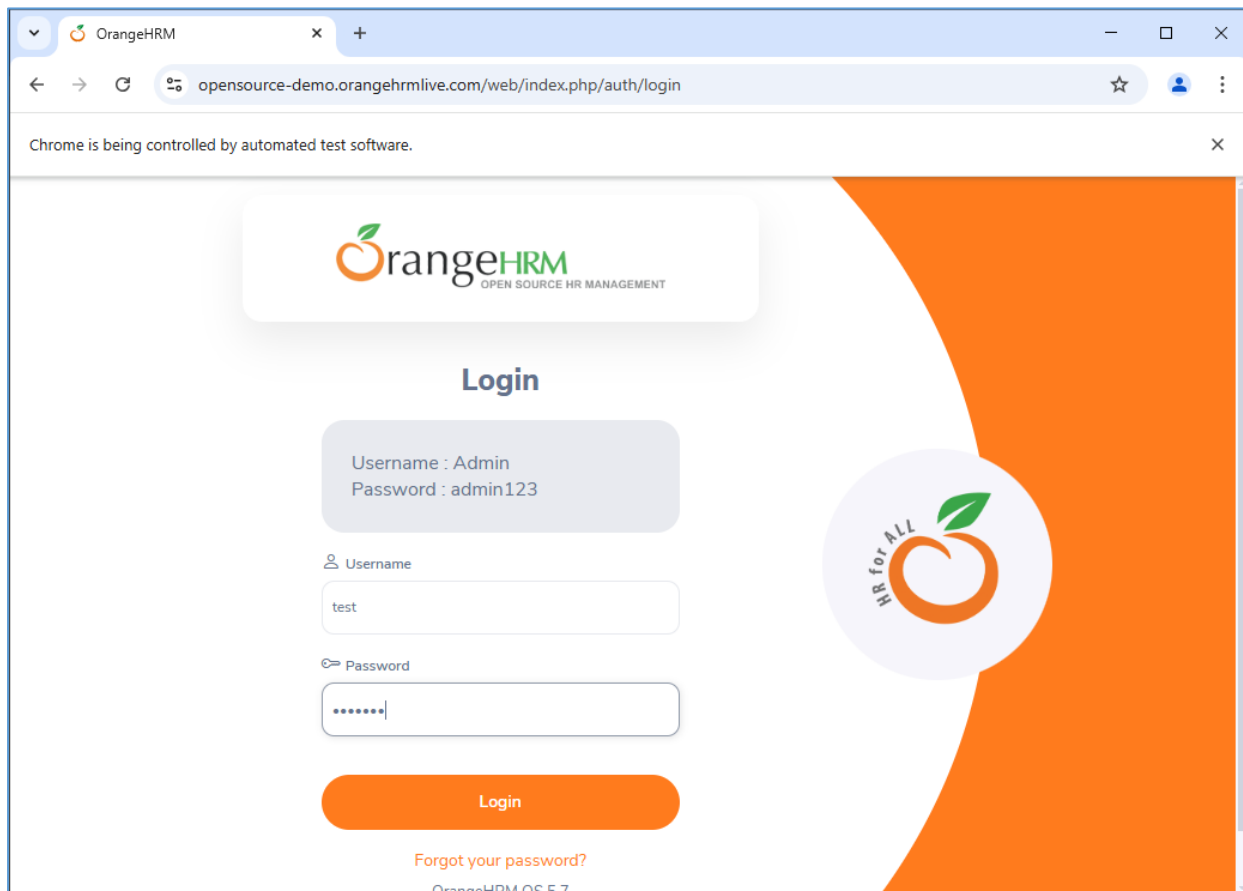


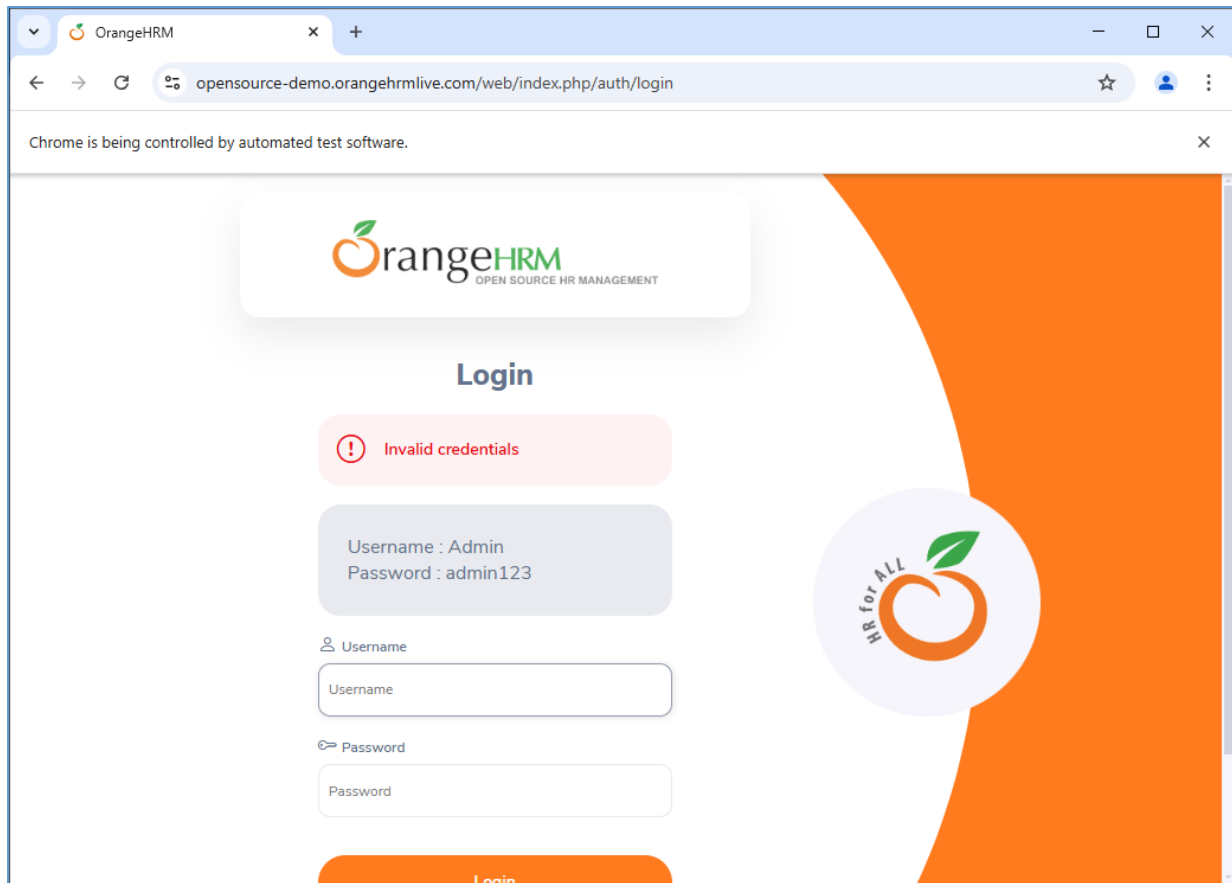
LOGIN4:



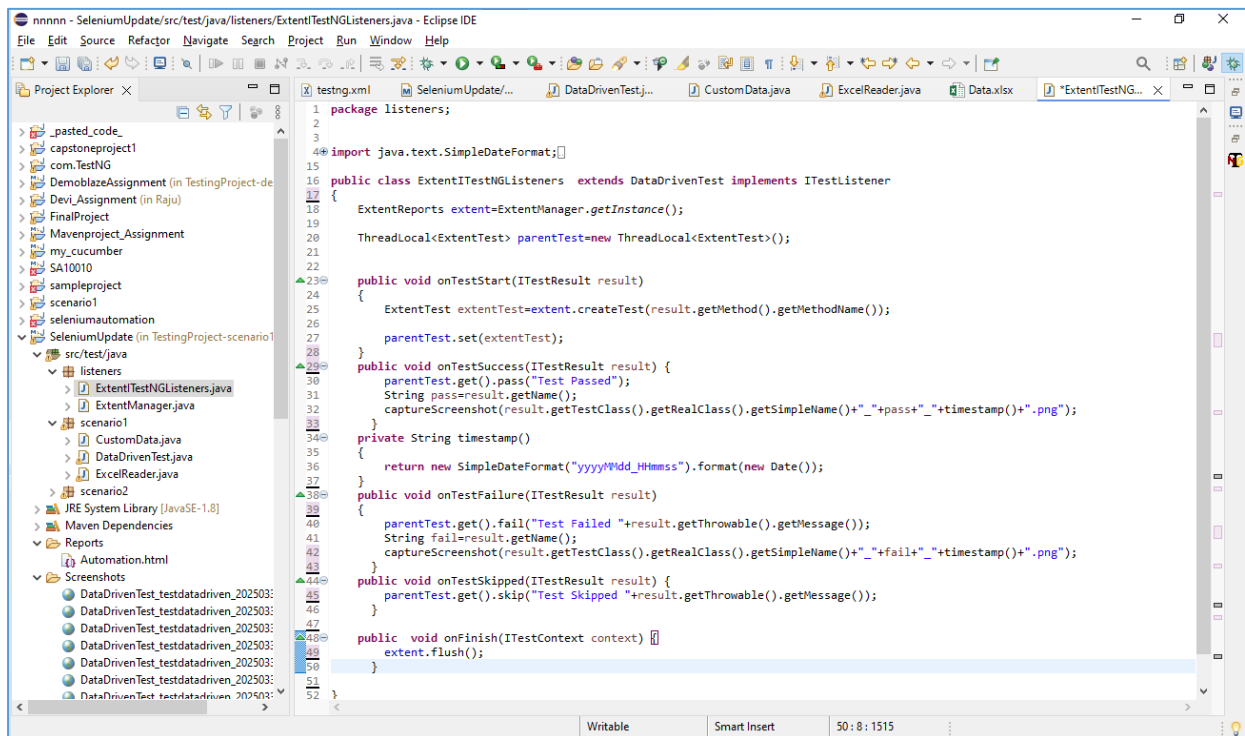


LOGIN5:





6. Generate Extent Report for the same.



nnnnn - SeleniumUpdate/src/test/java/listeners/ExtentManager.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer

- > _pasted_code
- > capstoneproject1
- > com.TestNG
- > DemoblazeAssignment (in TestingProject-de
- > Devi_Assignment (in Raju)
- > FinalProject
- > Mavenproject_Assignment
- > my_cucumber
- > SA10010
- > sampleproject
- > scenario1
- > seleniumautomation
- > SeleniumUpdate (in TestingProject-scenario1
- > src/test/java
 - > listeners
 - > Extent!TestNGListeners.java
 - > ExtentManager.java
 - > scenario1
 - > CustomData.java
 - > DataDrivenTest.java
 - > ExcelReader.java
 - > scenario2
- > JRE System Library [JavaSE-1.8]
- > Maven Dependencies
- > Reports
 - > Automation.html
- > Screenshots
 - > DataDrivenTest_testdatadriven_202503:
 - > DataDrivenTest_testdatadriven_202503:
 - > DataDrivenTest_testdatadriven_202503:
 - > DataDrivenTest_testdatadriven_202503:
 - > DataDrivenTest_testdatadriven_202503:
 - > DataDrivenTest_testdatadriven_202503:
 - > DataDrivenTest_testdatadriven_202503:

testng.xml SeleniumUpd... DataDrivenT... ExcelReader... Data.xlsx *Extent!Tes... ExtentManag...

```
1 package listeners;
2
3 import com.aventstack.extentreports.ExtentReports;
4
5 public class ExtentManager
6 {
7     public static ExtentReports extent;
8
9     public static ExtentReports getInstance()
10    {
11        if(extent==null)
12        {
13            extent=createInstance();
14            return extent;
15        }
16        else
17        {
18            return extent;
19        }
20    }
21
22    public static ExtentReports createInstance()
23    {
24        ExtentSparkReporter sparkReporter=new ExtentSparkReporter(System.getProperty("user.dir")+"/Reports/Automation.html");
25        sparkReporter.config().setTheme(Theme.DARK);
26        sparkReporter.config().setReportName("NewTestReport");
27        sparkReporter.config().setDocumentTitle("Sprint 1 Report");
28
29        ExtentReports extent=new ExtentReports();
30        extent.attachReporter(sparkReporter);
31
32        return extent;
33    }
34 }
35
36
37
38
39
40
41
42
43
44
```

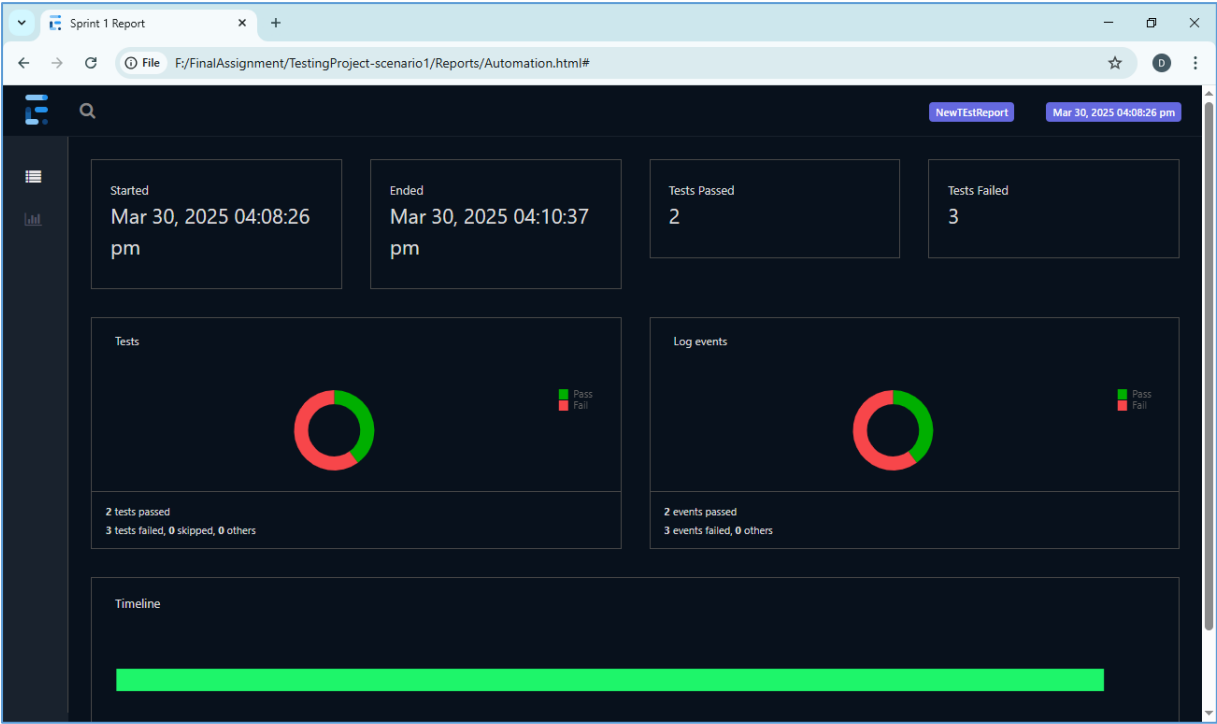
Writable Smart Insert 1:1:0

Sprint 1 Report

File F:/FinalAssignment/TestingProject-scenario1/Reports/Automation.html

NewTestReport Mar 30, 2025 04:08:26 pm

Tests	testdatadriven
testdatadriven 4:08:28 pm / 00:00:21:091	03.30.2025 4:08:28 pm 03.30.2025 4:08:49 pm 00:00:21:091 #test-id=1
testdatadriven 4:08:51 pm / 00:00:35:478	STATUS TIMESTAMP DETAILS
testdatadriven 4:09:26 pm / 00:00:30:206	Pass 4:08:49 pm Test Passed
testdatadriven 4:09:57 pm / 00:00:14:831	
testdatadriven 4:10:12 pm / 00:00:23:879	



TestNG Report

File F:/FinalAssignment/TestingProject-scenario1/test-output/emailable-report.html

Test	# Passed	# Skipped	# Retried	# Failed	Time (ms)	Included Groups	Excluded Groups
Suite							
Test	5	0	0	0	82,124		

Class	Method	Start	Time (ms)
Suite			
Test — passed			
scenario1.DataDrivenTest	testdatadriven	1743332921948	14878
	testdatadriven		
	testdatadriven		
	testdatadriven		
	testdatadriven		

Test

scenario1.DataDrivenTest#testdatadriven

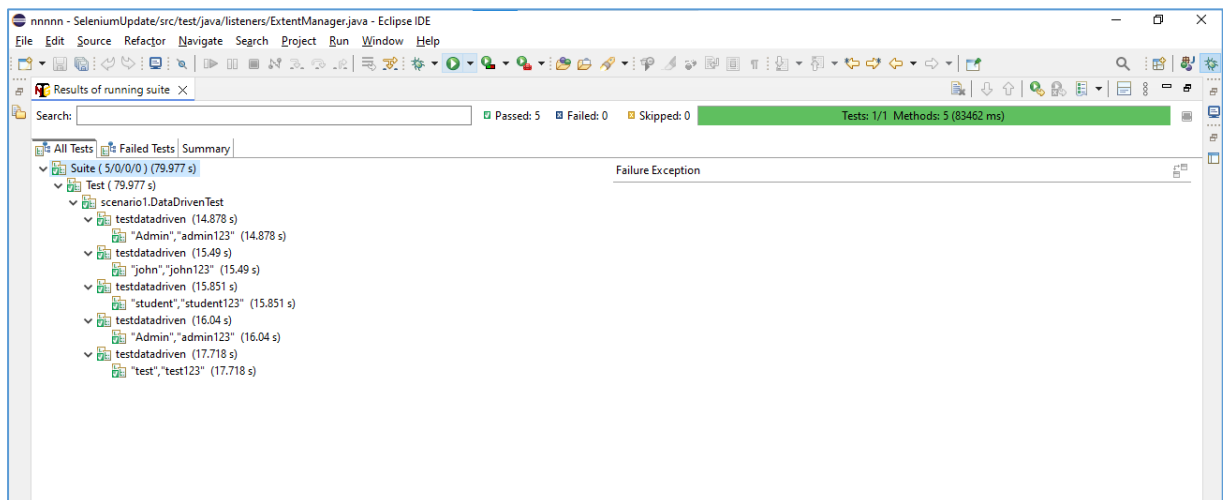
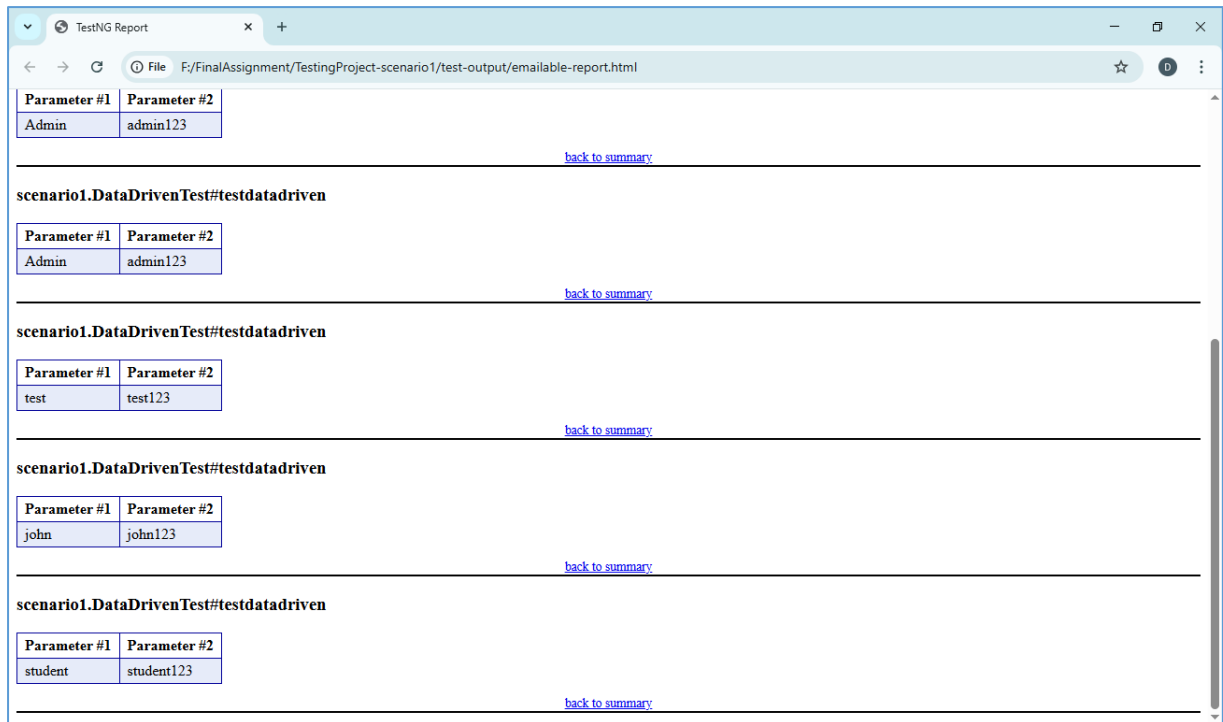
Parameter #1	Parameter #2
Admin	admin123

[back to summary](#)

scenario1.DataDrivenTest#testdatadriven

Parameter #1	Parameter #2
Admin	admin123

[back to summary](#)



Scenario 2: Create a Page Object Model for two pages LoginPage and Admin Page and write automation script for Login functionality and Admin search feature

Create a Page and Class for LoginPage and automate the functionality of OrangeHRM login for Valid Test Data: (username: Admin and Password: admin123).

Create a Page and class for Admin where you can prepare 4 important test cases

- 1. Create testcase to get all 12 options from left side menu and print the count which should be 12 from that list click on Admin then Admin page get open.**
- 2. Create test case for search For Existing Employee searchByUsername() : here send username Admin to username text box and click on search button and display total record found and refresh page.**
- 3. Create test case for search For Existing Employee searchByUserRole() : here automate dropdown and select Role Admin and click on search button and display total record found and refresh page.**
- 4. Create test case for search For Existing Employee searchByUserStatus() : here automate dropdown and select status Enabled or Disabled then click on search button and display total record found.**

CODE:

Login_Page.java:

```
package scenario2;
```

```
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.WebElement;  
import org.openqa.selenium.support.FindBy;  
import org.openqa.selenium.support.PageFactory;  
import org.testng.annotations.Test;
```

```
public class Login_Page {  
    WebDriver driver;  
  
    public Login_Page(WebDriver d) {  
        this.driver = d;
```

```
        PageFactory.initElements(driver, this);
    }

    @FindBy(name = "username")
    private WebElement username;
    @FindBy(name = "password")
    private WebElement password;
    @FindBy(xpath = "//button[@type=\"submit\"]")
    private WebElement submit;

    public WebElement getUsername() {
        return username;
    }
    public WebElement getPassword() {
        return password;
    }
    public WebElement getSubmit() {
        return submit;
    }

    @Test
    public void login(String un, String ps) {
        getUsername().sendKeys(un);
        getPassword().sendKeys(ps);
        try {
            Thread.sleep(2000);
        } catch (InterruptedException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        }
        getSubmit().submit();
        System.out.println("Login Successful");
    }
}
```

```
}
```

Admin_Page.java:

```
package scenario2;
```

```
import java.util.List;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
```

```
import org.openqa.selenium.support.FindBy;
```

```
import org.openqa.selenium.support.PageFactory;
```

```
import org.testng.Assert;
```

```
import org.testng.annotations.Test;
```

```
public class Admin_Page {
```

```
    private WebDriver driver;
```

```
    public Admin_Page(WebDriver d) {
```

```
        this.driver = d;
```

```
        PageFactory.initElements(driver, this);
```

```
    }
```

```
    @FindBy(xpath = "//ul[@class=\"oxd-main-menu\"]/li/a")
```

```
    private List<WebElement> menu_items;
```

```
    @FindBy(linkText = "Admin")
```

```
    private WebElement admin_link;
```

```
    @FindBy(xpath = "//div[@class=\"oxd-input-group oxd-input-field-bottom-space\"]/div[2]/input")
```

```
    private WebElement username;
```

```

    @FindBy(xpath = "//button[@type=\"submit\"]")
    private WebElement search_button;
    @FindBy(xpath = "//div[@class=\"orangehrm-horizontal-
padding orangehrm-vertical-padding\"]//span")
    private WebElement records_found;
    @FindBy(xpath = "//i[@class=\"oxd-icon bi-caret-down-fill
oxd-select-text--arrow\"]")
    private List<WebElement> drop_down;
    @FindBy(xpath = "//div[@role=\"listbox\"]/div/span")
    private List<WebElement> drop_down_items;
    @FindBy(xpath = "//div[@tabindex=\"0\"]")
    private List<WebElement> user_status_items;

    public List<WebElement> getUser_status_items() {
        return user_status_items;
    }
    public List<WebElement> getDrop_down() {
        return drop_down;
    }
    public List<WebElement> getdrop_down_items() {
        return drop_down_items;
    }
    public WebElement getRecords_found() {
        return records_found;
    }
    public WebElement getUsername() {
        return username;
    }
    public WebElement getSearch_button() {

```

```
        return search_button;
    }
    public List<WebElement> getMenu_items() {
        return menu_items;
    }
    public WebElement getAdmin_link() {
        return admin_link;
    }
    @Test
    public void menu_options() {
        System.out.println("Total number of Menu items:" +
            getMenu_items().size());
        getAdmin_link().click();
        String url = driver.getCurrentUrl();
        Assert.assertTrue(url.contains("admin"), "Url does not
match");
        System.out.println("Logged into Admin Page");
    }
    @Test
    public void searchByUsername(String un) {
        getUsername().sendKeys(un);
        getSearch_button().click();
        try {
            Thread.sleep(3000);
        } catch (InterruptedException e) {
            // TODO Auto-generated catch block
            e.printStackTrace();
        }
        System.out.println("Total records With username:" + un +
```

```
" is:" + getRecords_found().getText());
driver.navigate().refresh();
}
@Test
public void searchByUserRole() {
    getDrop_down().get(0).click();
    for (WebElement i : getdrop_down_items()) {
        if (i.getText().contains("Admin")) {
            i.click();
            break;
        }
    }
    getSearch_button().click();
    try {
        Thread.sleep(3000);
    } catch (InterruptedException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    }
    System.out.println("Total records with User Role Admin:"
+ getRecords_found().getText());
    driver.navigate().refresh();
}
public void searchByUserStatus() {
    try {
        Thread.sleep(3000);
    } catch (InterruptedException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    }
}
```

```

    }
    getDrop_down().get(1).click();
    for (WebElement i : getdrop_down_items()) {
        if (i.getText().contains("Enabled")) {
            i.click();
            break;
        }
    }
    getSearch_button().click();
    try {
        Thread.sleep(3000);
    } catch (InterruptedException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    }
    System.out.println("Total Records enabled:" +
        getRecords_found().getText());
}
}

```

POM_Client.java:

```
package scenario2;
```

```

import org.testng.annotations.Test;
import java.time.Duration;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.annotations.AfterTest;

```



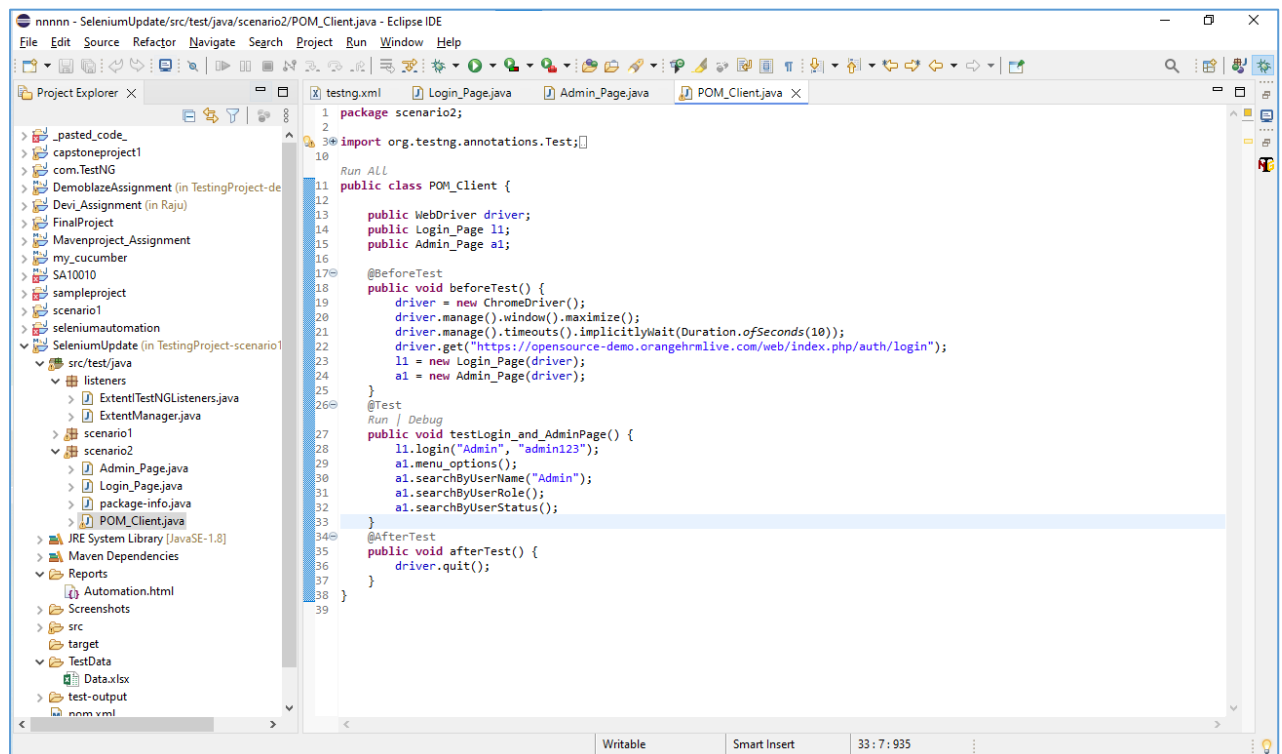
```
import org.testng.annotations.BeforeTest;
import org.testng.annotations.Test;
public class POM_Client {

    public WebDriver driver;
    public Login_Page l1;
    public Admin_Page a1;

    @BeforeTest
    public void beforeTest() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();

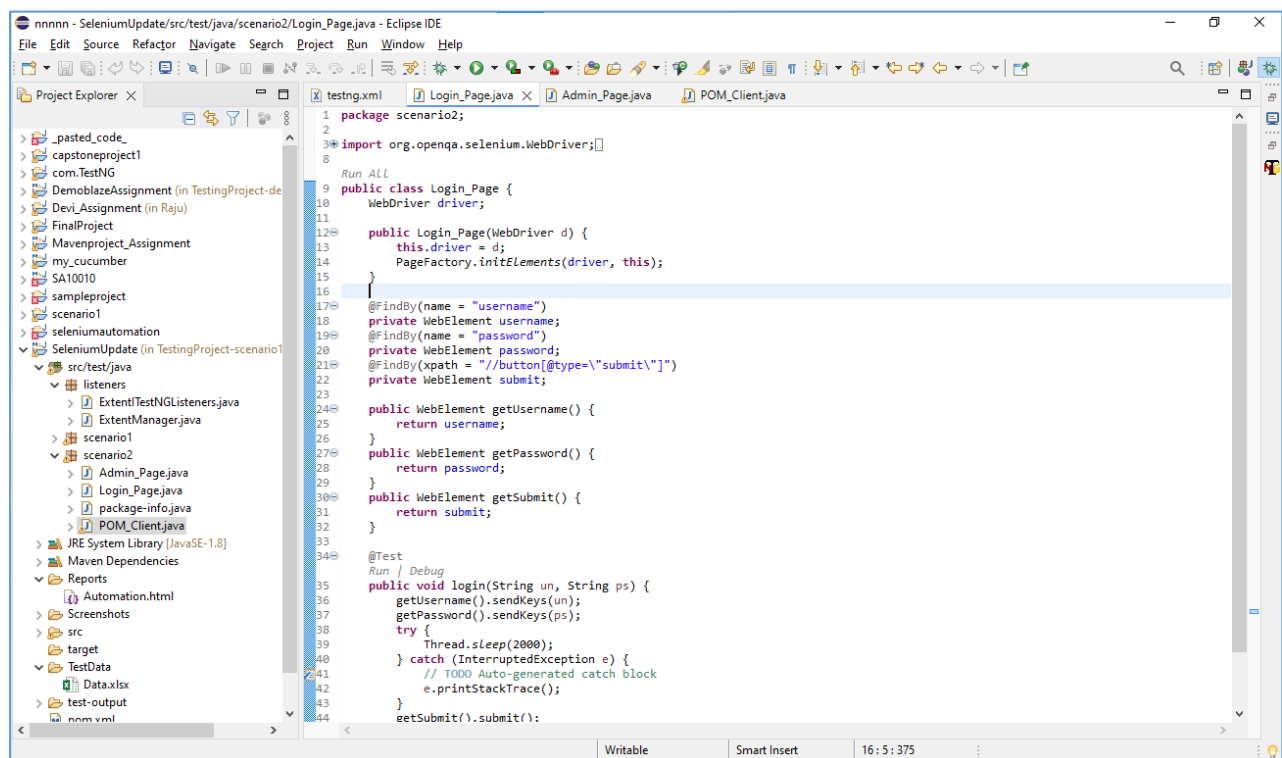
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
        driver.get("https://opensource-
demo.orangehrmlive.com/web/index.php/auth/login");
        l1 = new Login_Page(driver);
        a1 = new Admin_Page(driver);
    }
    @Test
    public void testLogin_and_AdminPage() {
        l1.login("Admin", "admin123");
        a1.menu_options();
        a1.searchByUsername("Admin");
        a1.searchByUserRole();
        a1.searchByUserStatus();
    }
    @AfterTest
    public void afterTest() {
        driver.quit();
    }
}
```

Create a Page and Class for LoginPage and automate the functionality of OrangeHRM login for Valid Test Data: (username: Admin and Password: admin123).



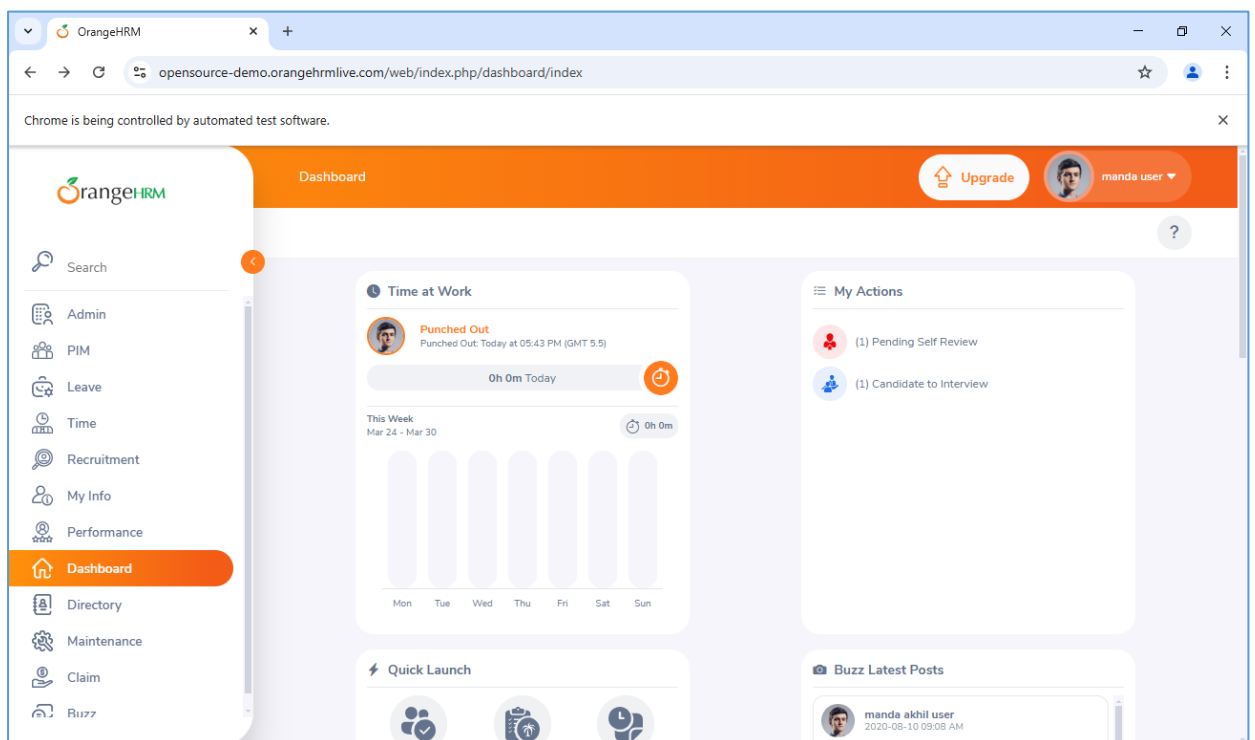
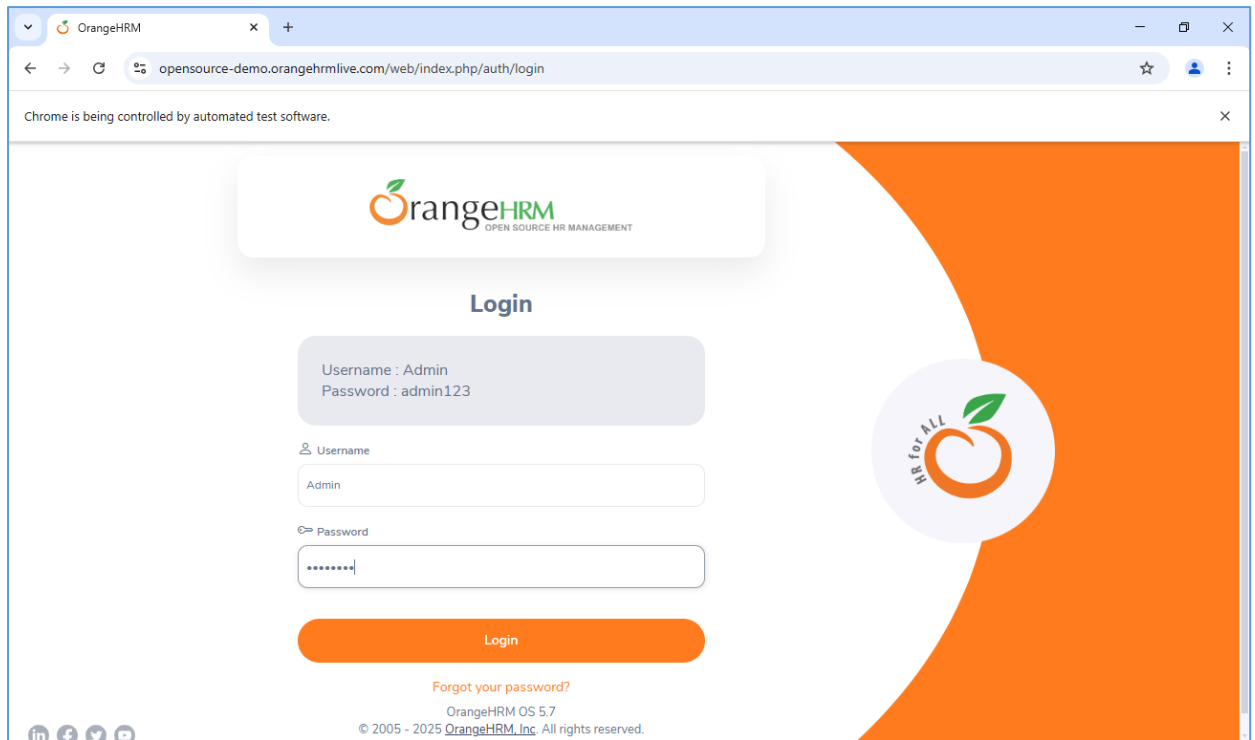
The screenshot shows the Eclipse IDE with the file `POM_Client.java` open. The Project Explorer on the left shows the project structure, including `src/test/java` and `scenario2`. The main editor displays the following Java code:

```
1 package scenario2;
2
3 import org.testng.annotations.Test;
4
5 Run ALL
6
7 public class POM_Client {
8
9     public WebDriver driver;
10    public Login_Page l1;
11    public Admin_Page a1;
12
13    @BeforeTest
14    public void beforeTest() {
15        driver = new ChromeDriver();
16        driver.manage().window().maximize();
17        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
18        driver.get("https://opensource-demo.orangehrmlive.com/web/index.php/auth/login");
19        l1 = new Login_Page(driver);
20        a1 = new Admin_Page(driver);
21    }
22
23    @Test
24    Run / Debug
25    public void testLogin_and_AdminPage() {
26        l1.login("Admin", "admin123");
27        a1.menu_options();
28        a1.searchByUserName("Admin");
29        a1.searchByUserRole();
30        a1.searchByUserStatus();
31    }
32
33    @AfterTest
34    public void afterTest() {
35        driver.quit();
36    }
37
38 }
39
```

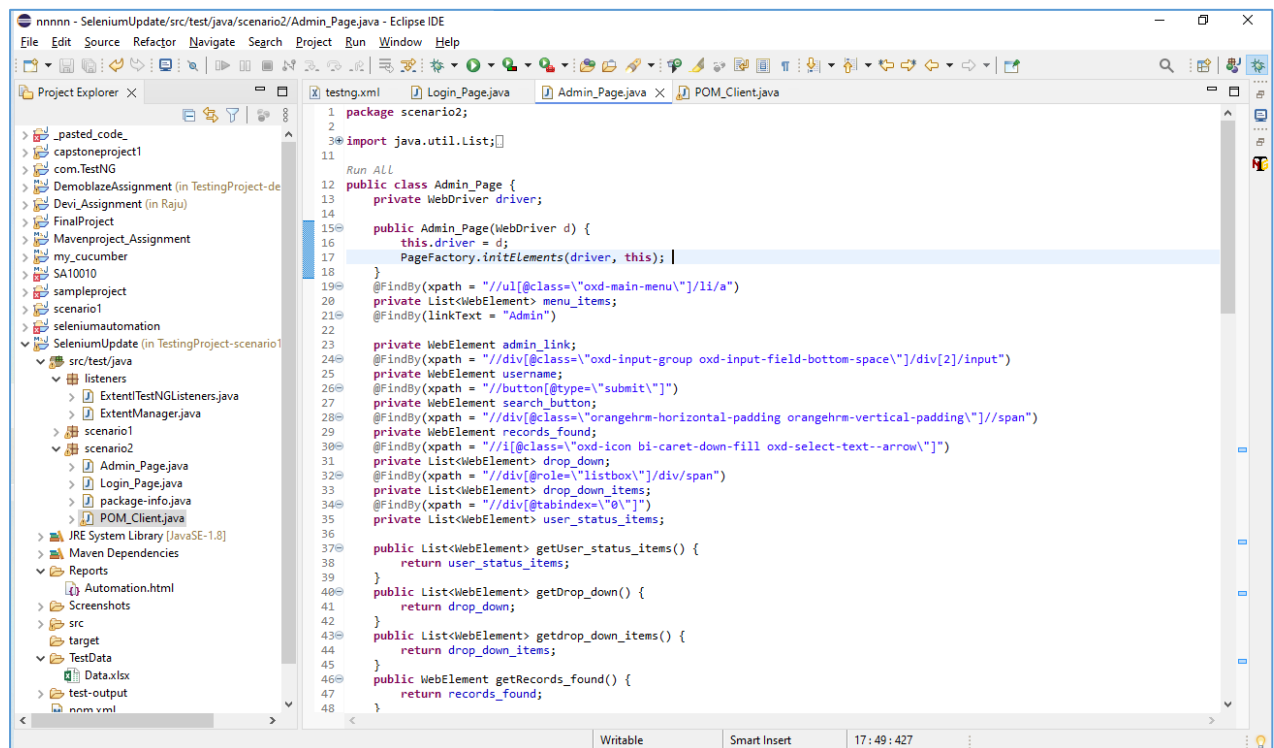


The screenshot shows the Eclipse IDE with the file `Login_Page.java` open. The Project Explorer on the left shows the project structure, including `src/test/java` and `scenario2`. The main editor displays the following Java code:

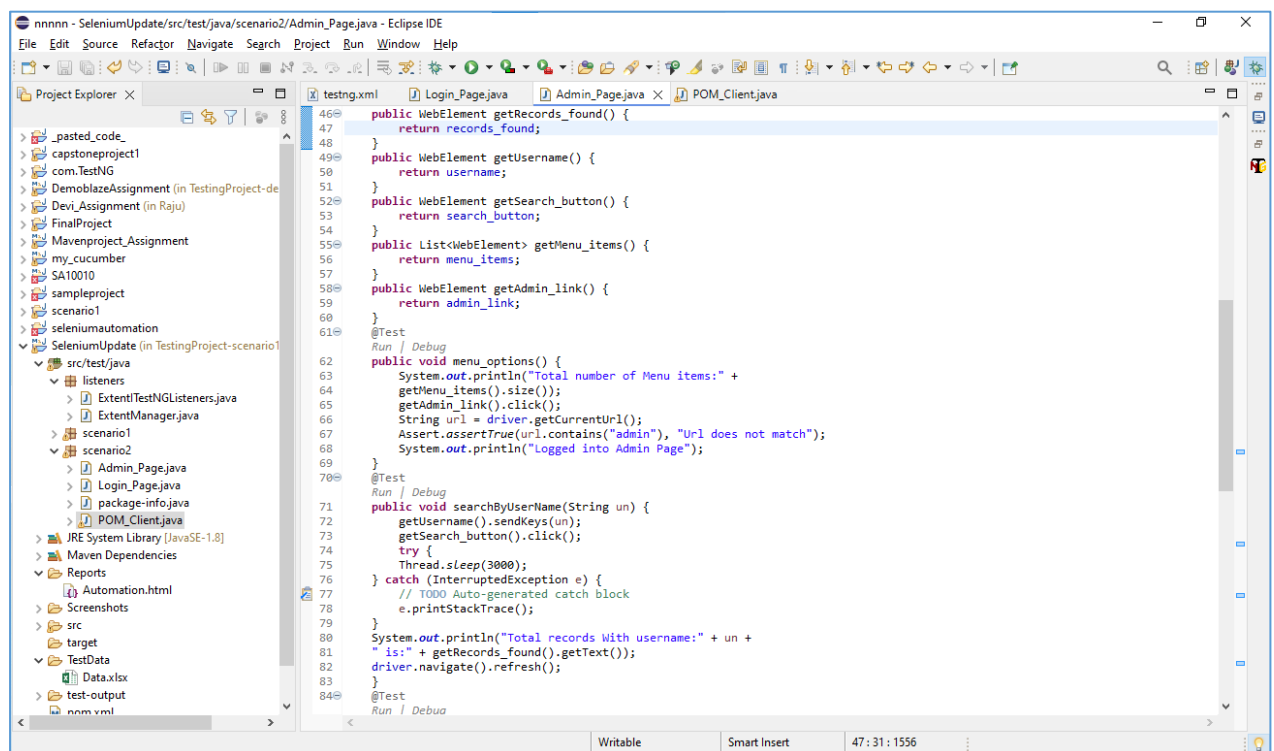
```
1 package scenario2;
2
3 import org.openqa.selenium.WebDriver;
4
5 Run ALL
6
7 public class Login_Page {
8     WebDriver driver;
9
10    public Login_Page(WebDriver d) {
11        this.driver = d;
12        PageFactory.initElements(driver, this);
13    }
14
15    @FindBy(name = "username")
16    private WebElement username;
17    @FindBy(name = "password")
18    private WebElement password;
19    @FindBy(xpath = "//button[@type='submit']")
20    private WebElement submit;
21
22    public WebElement getUsername() {
23        return username;
24    }
25
26    public WebElement getPassword() {
27        return password;
28    }
29
30    public WebElement getSubmit() {
31        return submit;
32    }
33
34    @Test
35    Run / Debug
36    public void login(String un, String ps) {
37        getUsername().sendKeys(un);
38        getPassword().sendKeys(ps);
39        try {
40            Thread.sleep(2000);
41        } catch (InterruptedException e) {
42            // TODO Auto-generated catch block
43            e.printStackTrace();
44        }
45        getSubmit().submit();
46    }
47
48 }
```



Create testcase to get all 12 options from left side menu and print the count which should be 12 from that list click on Admin then Admin page get open

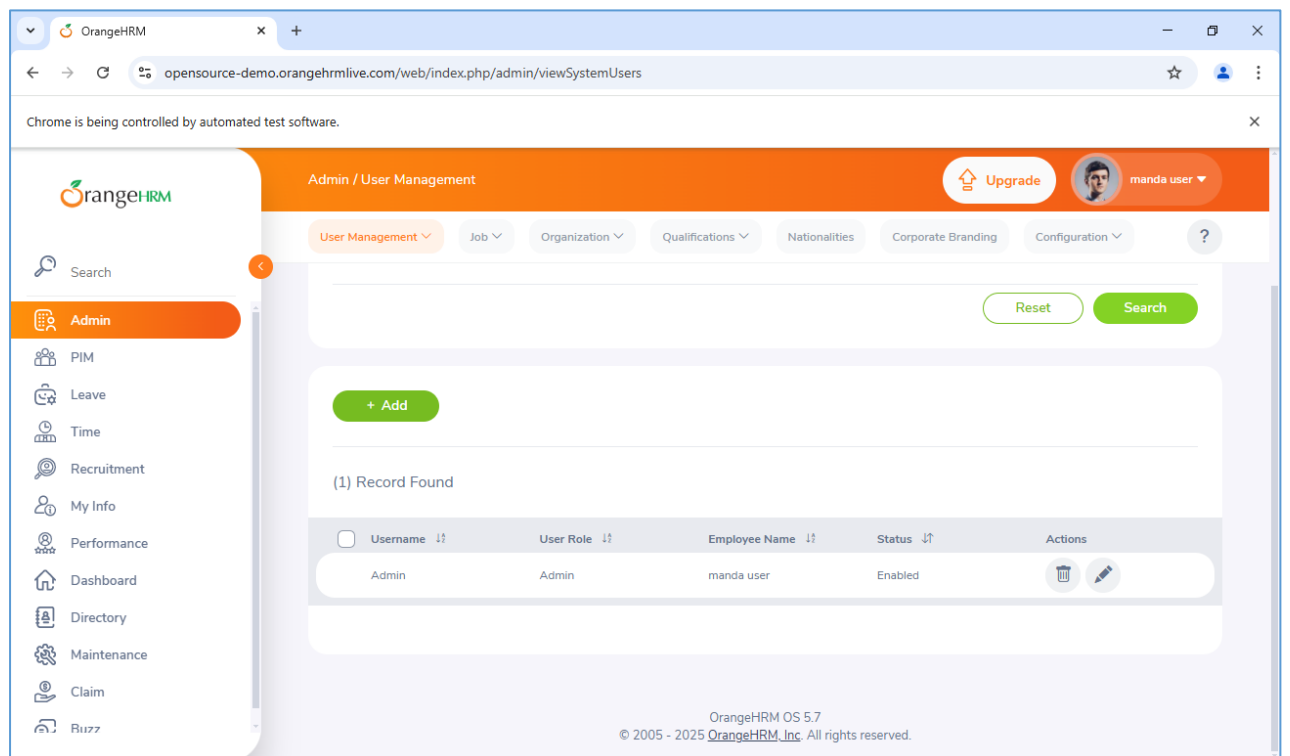
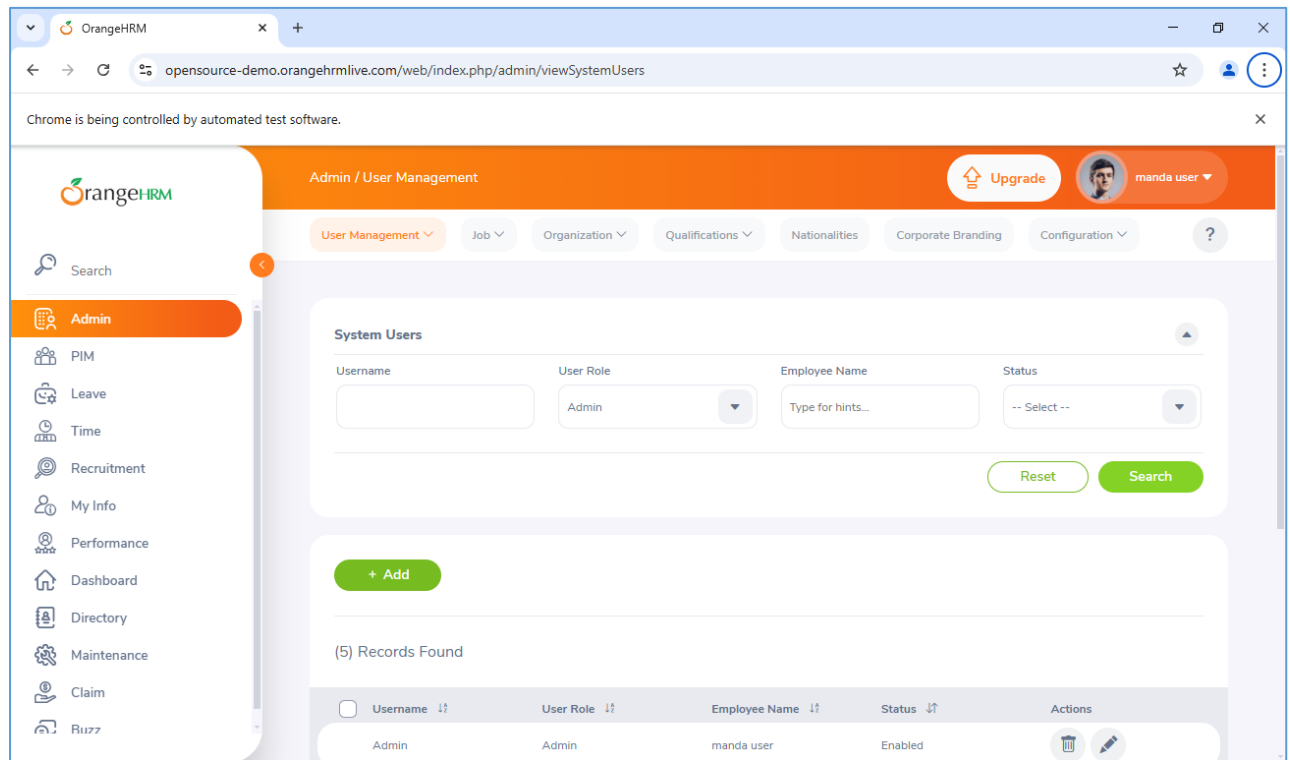


```
1 package scenario2;
2
3 import java.util.List;
4
5
6
7
8
9
10
11
12 public class Admin_Page {
13     private WebDriver driver;
14
15     public Admin_Page(WebDriver d) {
16         this.driver = d;
17         PageFactory.initElements(driver, this);
18     }
19     @FindBy(xpath = "//ul[@class='oxd-main-menu']/li/a")
20     private List<WebElement> menu_items;
21     @FindBy(linkText = "Admin")
22
23     private WebElement admin_link;
24     @FindBy(xpath = "//div[@class='oxd-input-group oxd-input-field-bottom-space']/div[2]/input")
25     private WebElement username;
26     @FindBy(xpath = "//button[@type='submit']")
27     private WebElement search_button;
28     @FindBy(xpath = "//div[@class='orangehrm-horizontal-padding orangehrm-vertical-padding']/span")
29     private WebElement records_found;
30     @FindBy(xpath = "//i[@class='oxd-icon bi-caret-down-fill oxd-select-text--arrow']")
31     private List<WebElement> drop_down;
32     @FindBy(xpath = "//div[@role='listbox']/div/span")
33     private List<WebElement> drop_down_items;
34     @FindBy(xpath = "//div[@tabindex='0']")
35     private List<WebElement> user_status_items;
36
37     public List<WebElement> getUser_status_items() {
38         return user_status_items;
39     }
40     public List<WebElement> getDrop_down() {
41         return drop_down;
42     }
43     public List<WebElement> getdrop_down_items() {
44         return drop_down_items;
45     }
46     public WebElement getRecords_found() {
47         return records_found;
48     }
```

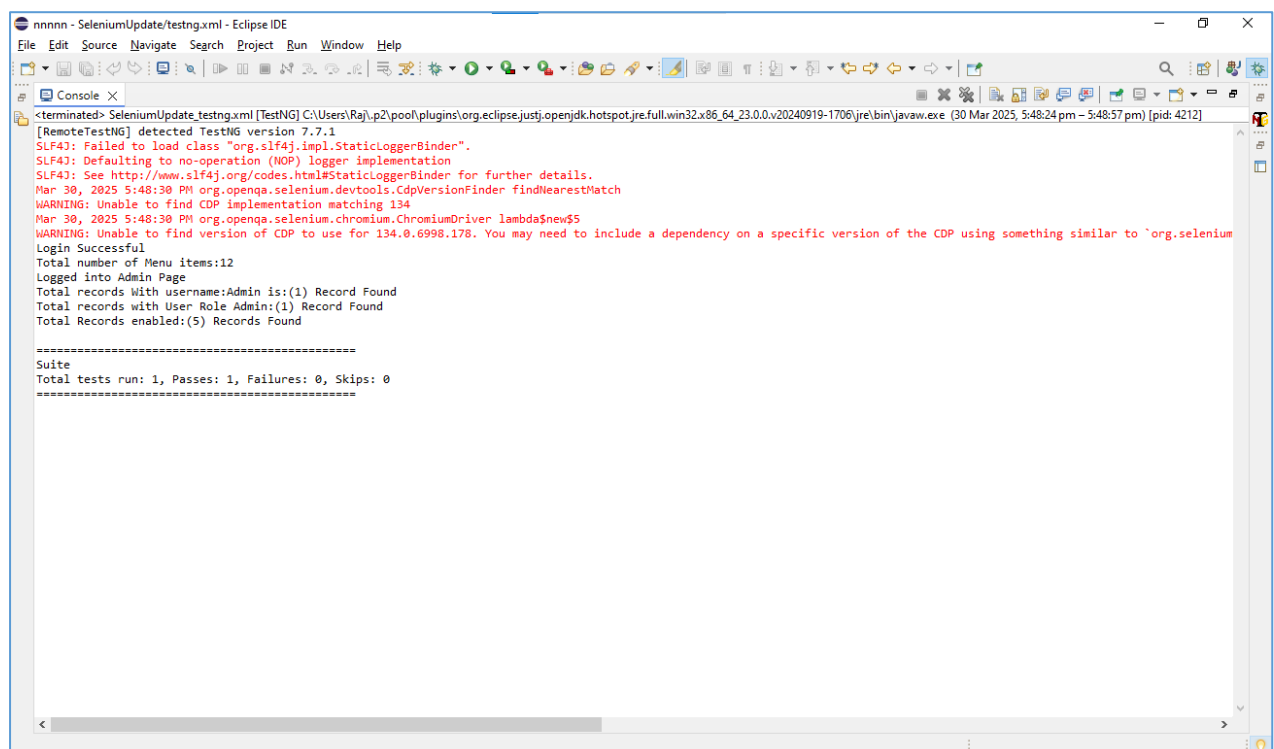
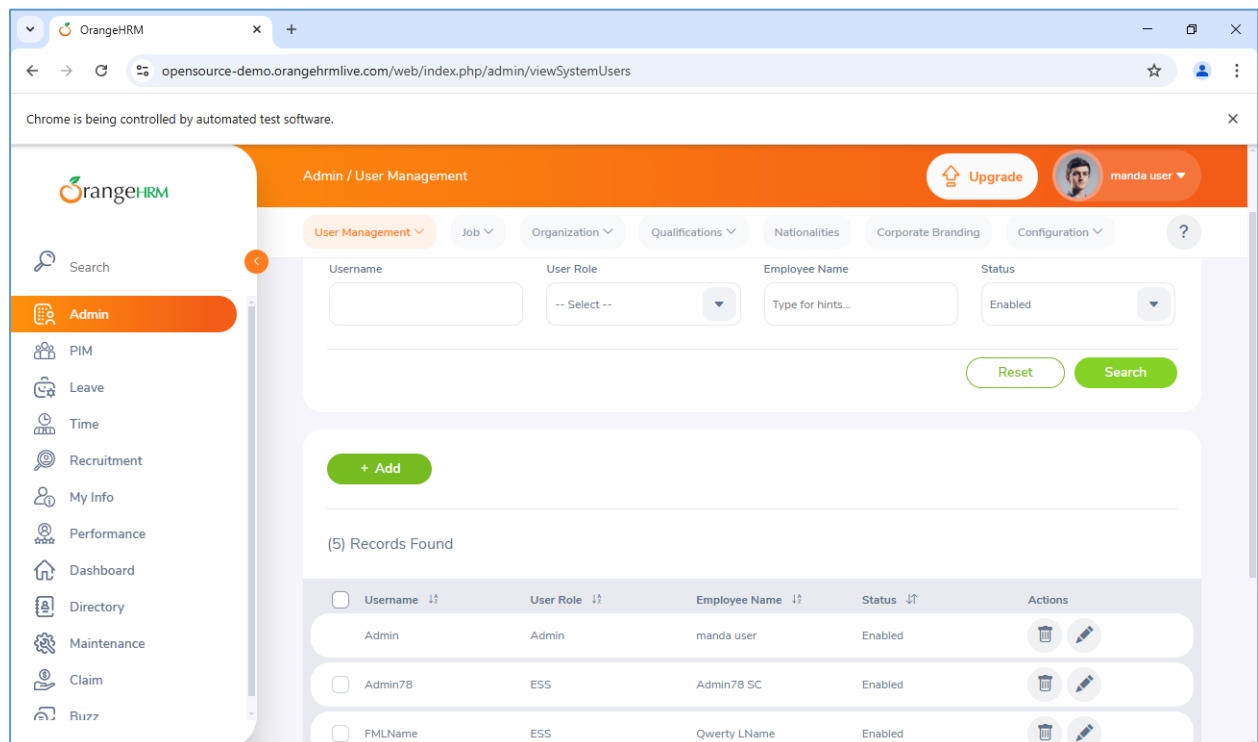


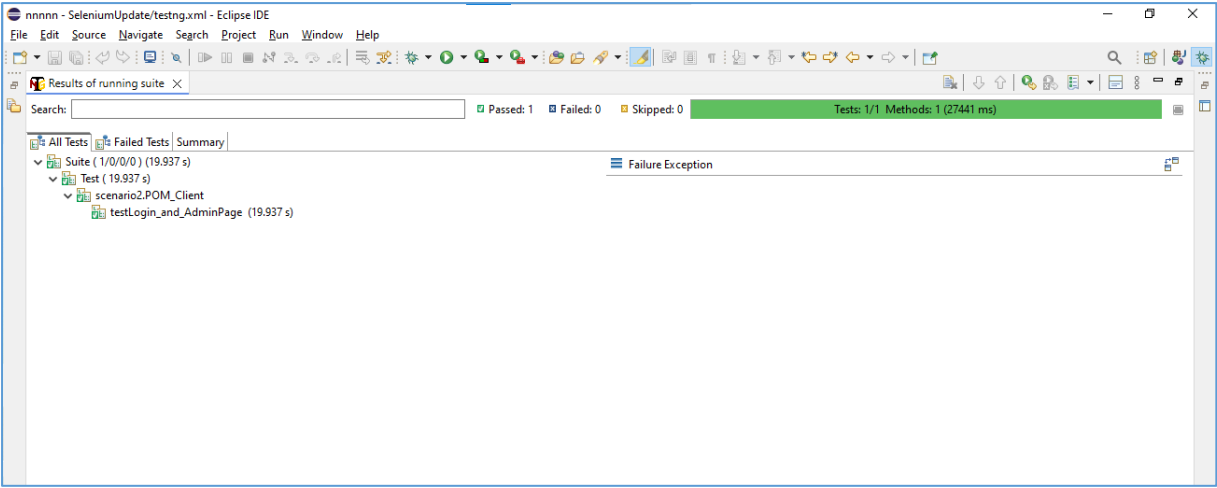
```
46 public WebElement getRecords_found() {
47     return records_found;
48 }
49 public WebElement getUsername() {
50     return username;
51 }
52 public WebElement getSearch_button() {
53     return search_button;
54 }
55 public List<WebElement> getMenu_items() {
56     return menu_items;
57 }
58 public WebElement getAdmin_link() {
59     return admin_link;
60 }
61 @Test
62 Run | Debug
63 public void menu_options() {
64     System.out.println("Total number of Menu items:" +
65         getMenu_items().size());
66     getAdmin_link().click();
67     String url = driver.getCurrentUrl();
68     Assert.assertTrue(url.contains("admin"), "Url does not match");
69     System.out.println("Logged into Admin Page");
70 }
71 @Test
72 Run | Debug
73 public void searchByUsername(String un) {
74     getUsername().sendKeys(un);
75     getSearch_button().click();
76     try {
77         Thread.sleep(3000);
78     } catch (InterruptedException e) {
79         // T000 Auto-generated catch block
80         e.printStackTrace();
81     }
82     System.out.println("Total records With username:" + un +
83         " is:" + getRecords_found().getText());
84     driver.navigate().refresh();
85 }
```


Create test case for search For Existing Employee searchByUserRole() : here automate dropdown and select Role Admin and click on search button and display total record found and refresh page.



Create test case for search For Existing Employee searchByUserStatus() : here automate dropdown and select status Enabled or Disabled then click on search button and display total record found.





← → ↺ File F:/FinalAssignment/TestingProject-scenario1/test-output/emailable-report.html ☆ ⓘ ⋮

Test	# Passed	# Skipped	# Retried	# Failed	Time (ms)	Included Groups	Excluded Groups
Suite							
Test	1	0	0	0	27,109		

Class	Method	Start	Time (ms)
Suite			
Test — passed			
scenario2.POM_Client	testLogin_and_AdminPage	1743337458280	19937

Test

scenario2.POM_Client#testLogin_and_AdminPage

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